

Conditional Risk Factor Investing

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Abstract

There is no consensus on how to apply a factor investing strategy. In this empirical work, we define factor investing as a strategy that replicates the behaviour of a significant risk factor in an asset pricing model, amplifying (mitigating) its effect by taking positions in individual assets with high (low) bet as if the premium associated with the factor was positive (negative). For a sample composed of US stocks and stocks from developed markets, excluding US stocks, for 2002-2023, we find that the conditional strategy performs better in terms of return risk and accrued value than the eight factors analysed.

Keywords: Asset pricing model, factor model, conditional investing.

JEL: C50, G11, G12

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Abstract

There is no consensus on how to apply a factor investing strategy. In this empirical work, we define factor investing as a strategy that replicates the behaviour of a significant risk factor in an asset pricing model, amplifying (mitigating) its effect by taking positions in individual assets with high (low) betas if the premium associated with the factor was positive (negative). For a sample composed of US stocks and stocks from developed markets, excluding US stocks, for 2002-2023, we find that the conditional strategy performs better in terms of return risk and accrued value than the eight factors analysed.

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1. Background

Factor models have a long history, with a wide range of theoretical models and practical applications for evaluating the performance of portfolio managers, creating investment strategies, assessing return anomalies, predicting returns, and constructing portfolios. The most prominent examples of theoretical models are asset pricing models, such as the capital asset pricing model (CAPM) and arbitrage pricing theory (APT), as well as the Fama–French factor model. The extensive research of Fama & French (1992), Fama & French (1993), Fama & French (1995), Fama & French (1996), Fama & French (1998) and Fama & French (2015) expanded the initial CAPM model of Tsuji (1964), Lintner (1965) and Black (1972) from a factor analysis perspective to include additional factors to explain cross-sectional variation in stock returns.

The factors most commonly used in empirical studies are: size, growth, profitability and investment (Fama-French); momentum, Carhart (1997); investment quality factors, Asness et al. (2019); and zero-beta by Frazzini & Pedersen (2014).

Moreover, during the last 20 years, attempts have been made to determine new possible explanatory factors for the behavior of asset returns, among others Pastor & Stambaugh (2003) included liquidity and Ang et al. (2006) analysed the asymmetry of risk. Thus, there is an extensive literature that has studied these factors with contrary results, in some cases corroborating their validity (e.g., Grinold (1993), Clare et al. (1998), Faff (2003), Nichol & Dowling (2014)), and in others questioning it (among others, Kothari et al. (1995), Chiah et al. (2016)).

Additionally, it is found that the results are conditioned, among other factors, by the geographical scope of the study; however, in global terms, the results are inconclusive (Cakici & Zaremba (2015), Fama & French (2017), Schmidt et al. (2019), Cakici & Zaremba (2021)). Moreover, the factors of size and profitability have been explored more in global markets (Asness et al. (2013), Cakici et al. (2013), Cakici & Tan (2014), Hanauer & Linhart (2015), among others), than the factors of profitability (Chen et al. (2018)) or investment (Watanabe et al. (2013), Titman et al. (2013)). Thus, for example, for a sample of 23 countries over the

25 period 1990-2019, Cakici & Zaremba (2021) reported that the explanatory power of the value, profitability and investment factors is generally low, although the value premium in Asia and the investment premium in Europe are significant. In addition, a limitation recognized by the authors is that the momentum factor has not been considered.

Thus, we note that while the literature has extensively analysed the explanatory power of factors (for
30 example, Harvey et al. (2016), Blitz et al. (2020) and Enow (2023)), there are few and inconclusive works that use factors to guide investment based on the relative importance of each factor; for example, there are studies that employ these factors in the analysis of the performance of industrial or sector portfolios (see Tsuji (2012), Sarwar et al. (2018) and Laopodis (2023)); however, in general, few studies on factor-based investment exist. Beck et al. (2016) reported that momentum and market betas were more robust than
35 size, although the results were conditioned by trading costs. Additionally, Bender (2023) concluded that the value factor has performed worse than the market and that the results of the momentum and quality factors are very volatile.

In the literature, factor investing has been described as a “new paradigm” of long-term investing (see for example, Ang (2014), Cazalet & Roncalli (2014), Jurczenko et al. (2015) and Brière & Szafarz (2017)).
40 Thus, factor-based investment strategies or factor investing have been gaining prominence, especially since the 2008 financial crisis¹, both in the professional (see Ang et al. (2009)) and academic spheres (Cazalet & Roncalli (2014)). As Asness et al. (2015) noted, institutional investors have since turned their interest to strategies that offer alternative sources of returns that are less correlated with traditional market returns. All of this prompted a renewed interest in factor investing strategies, defined by Asness et al. (2015) as
45 systematic simultaneous long-short strategies that offer positive long-term average returns across markets and asset classes with low global correlations with long-only portfolios. The objective is to combine factors to improve the performance of portfolios in the long term and, in any case, smart beta portfolios refer to investment strategies that are not governed by the traditional method of weighting components by market capitalization (see Roncalli (2014), Monga et al. (2022)). Several studies have shown that factor-constructed
50 portfolios have outperformed the market over the long run (Dimson et al. (2017), Bessler et al. (2021)). However, as Ang (2014) noted “*risk premiums don’t come for free*”, factors can underperform in the short run and are cyclical so they require dynamic strategies that constantly involve adjustments of the weights in the portfolio

Currently, the results of any empirical work in this field are fundamental, not only because of the economic
55 implications but also because they shed some light on the multitude of works that incorporate new factors (see for example, Blitz et al. (2021), Blitz (2023)), which, makes it extremely difficult to determine the relevant factors of the total number of variables studied, for example, Harvey & Liu (2019) document more

¹An important impetus also came from the publication of the report by Ang et al. (2009) at the request of the Norwegian sovereign wealth fund which concluded that approximately 70% of the fund’s portfolio returns could be explained by exposure to systematic factors.

than 400 factors published only in scientific journals; hence Cochrane (2011) has coined the term “zoo of new factors” to refer to their exponential growth. Some authors consider that such factor inflation is actually a questionable development because most of the proposed factors are likely to be spurious/nonconsistent and the result of data mining, multiple testing or p-hacking (Feng et al. (2020), Harvey et al. (2016), Harvey (2017), Hou et al. (2019)). In this context, Ang et al. (2009) and Ang (2014) noted that factors to invest in should meet the following criteria: i) be justified by academic research, ii) have experienced significant risk premia that are expected to persist in the future, iii) have proven history of returns in bad times, and iv) be implementable in liquid instruments, v) have a high risk premium that is expected to persist in the future, vi) have a proven history of returns in bad times, and vii) be implementable in liquid instruments.

Within the empirical work on factor investing there seems to be a certain consensus on the differentiation of investor strategy between the long and the short term. While in the first case factors show positive results, as shown by Bender et al. (2013a) and Bender et al. (2013b); in the short term returns are more volatile, and mixed strategies or combinations of factors are proposed to mitigate this effect (Bender et al. (2013a), Bender et al. (2013b), Cazalet & Roncalli (2014) and Roncalli (2014)).

A second complex element is related to the implementation of factor investing strategies. As Bender et al. (2013a) and Bender et al. (2013b) noted, the factor portfolios proposed in the literature do not consider the liquidity of the stocks included; their difficulty in taking short positions does not include transaction costs or the rebalancing is performed in shorter time frames than those usually employed by the financial industry.

As indicated by Cazalet & Roncalli (2014), so far some relevant facts on factor investing strategies lack empirical contrast:

- Common risk factors explain more variance than idiosyncratic risk factors in diversified portfolios.
- Some risk factors are more relevant than others are, which may imply higher returns.
- Risk premiums vary over time and revert to an average value in the long run.
- The outcome of factors that consist only of long positions (e.g. market) should be differentiated from those that include both long and short positions (e.g. zero-beta and momentum).
- The geographical effect of the factors is significant; therefore, the risk factors are not geographically homogeneous.

Moreover, Arnott et al. (2019), Arnott et al. (2023) and Aghassi et al. (2023) identify concepts and concerns that are not sufficiently resolved in the literature on factor investing:

- Avoid exaggerated expectations motivated by data mining and backtest overfitting. Additionally, Arnott et al. (2019) noted that factors exhibit worse behaviour and results after they are made public compared with their previous trend. This may be due to excess factor demand known as overcrowding (or simply crowding).

- Factors sometimes experience large drawdowns, prolonged periods of underperformance and high kurtosis, the effects of which are magnified by volatilities and time-varying correlations.
- Factor returns have thick tails and downside skewness, which is why it is necessary to analyse factor behaviour in different environments, as some are prone to crashes. The momentum factor for example exhibits large negative skewness and is prone to large crashes. Daniel & Moskowitz (2016) observe that momentum crashes during periods of high volatility, which usually occur during a market crash. Thus, the momentum factor and most factor returns depart from a normal distribution.

A relevant contribution is proposed by Kim (2022) who postulates the convenience of exploiting the momentum property (trending property) exhibited by the factors to propose a tactical factor allocation (TFA) scheme for the construction of multifactor portfolios. This paper analyses the factors of size, value, quality, investment and momentum on the basis of worldwide data² from French's Data Library. To implement the TFA, the five factors are initially equally weighted in the portfolio, but if the factor's performance is positive in the previous 12 months then its weight is increased³ from 1 to 1.5 when the portfolio is rebalanced. Otherwise, the weight would remain unchanged. Quarterly rebalancing and volatility scaling are employed in each portfolio based on daily returns over the last 6 months. The results reveal that the TFA outperforms an equally weighted (EW) scheme in a persistent manner over time and in the different markets analysed, considers different performance measures (for example, in US (developed ex-US) the TFA outperforms EW by 21.9% (19.3%) in terms of the information ratio, and while EW produces a maximum drawdown of 34.4% in the US, the TFA does so by 24.1%).

A topic that is receiving renewed interest relates to the iterative (or recursive) combination of factors. Bermejo et al. (2021) evaluate whether a systematic investment strategy based on value, profitability and momentum factors can generate a positive alpha and outperform benchmark strategies in the European equity market. To do so, they use a sample of 600 large cap companies in Europe between 1991 and 2019, excluding banks, insurers and REITs. The data are sourced from the FactSet database and include accounting and market price metrics. Value drivers are measured using Book-to-Market (BTM), Price-to-Earnings Ratio (PER), EV/EBIT and EV/EBITDA; profitability is assessed using Gross Profit to Assets (GPA) and Return on Capital (ROC); and momentum is measured using 12-month cumulative profitability, excluding the most recent to avoid reversal effects documented in Jegadeesh (1990) and Jegadeesh & Titman (1993). The methodology employs three stock selection approaches: pure strategies, which use a single factor; mixed strategies, which combine two factors (value with profitability/momentum) by averaging their rankings; and iterative (conditional) strategies, which apply sequential filters (first value, second profitability and third momentum). The analysis includes a total of 40 rankings, resulting from 8, 16 and 16 pure, mixed and

²The periods considered are: January 1967 to September 2020 for the US, April 1994 to September 2020 for the developed ex-US, May 1992 to September 2020 for Europe and May 1992 to September 2020 for the Asia-Pacific area (except Japan).

³The author indicates that, although the choice of the increase in weighting implies a certain arbitrariness, as long as it is reasonable, the chosen increase would have similar effects in terms of strategy performance.

conditional strategies, respectively⁴. Equally weighted portfolios are constructed in the paper, and only long positions are employed. The results show that factor-based strategies generate annualized returns of up to 17%, with iterative portfolios significantly outperforming pure and mixed portfolios. However, the returns of these strategies have decreased over time because of the increasing adoption of factor investing, revealing a factor crowding phenomenon documented in the literature (Linnainmaa & Roberts (2018), Arnott et al. (2019), Fama & French (2020) and Hua & Sun (2024)). The combination of value and momentum (P/E more Momentum) proved to be the most effective strategy, showing the highest alpha and cumulative return. These findings suggest that systematic factor-based investing can be a profitable strategy, but its effectiveness is subject to changes in the market and competition among investors.

One line of research on which there is an open debate in the factor investing literature is the comparison of multifactor portfolio construction techniques coined as top-down (also known as *portfolio blending* or *mixing*) and bottom-up (also known in the literature as *signal blending* or *integrating*). The top-down approach consists of two stages. In the first stage, single-factor portfolios are constructed. These single-factor portfolios are then combined into a multifactor portfolio by assigning each single factor portfolio the desired weighting. The bottom-up approach constructs the multifactor portfolio in a single step by selecting the desired number of stocks that, on average, have a high exposure to all risk factors for interest simultaneously.

Proponents of bottom-up construction include Clarke et al. (2016)⁵, Bender & Wang (2016)⁶ and Fitzgibbons et al. (2017)⁷. Other papers doubt that the bottom-up approach outperforms the top-down approach on a risk-adjusted basis. Leippold & Rueegg (2018) find, using a multiple hypothesis framework⁸, that

⁴Pure rankings (8): created using the 4 individual value metrics, 3 profitability metrics and 1 momentum metric. Mixed rankings (16): created by combining each value metric with profitability (4x3=12) and momentum (4x1=4). Conditional rankings (16), with two alternatives: a) created by selecting the first quintile of each value metric, then the top 50% (portfolios including the top 5% for each of the combinations considered) for each profitability metric and then the top 50% for momentum (4x3x1=12). Each portfolio includes 30 stocks; b) created by selecting the top quintile for each value metric and then the top 25% by momentum (4x1=4). They then select the top 25% to create comparable portfolios of 30 stocks for all conditional rankings.

⁵In this paper both techniques of portfolio construction with long positions are compared using data of 1000 US stocks during the period 1968-2015. The factors employed are low beta, size, value and momentum. The authors find that bottom-up construction portfolios achieved a Sharpe ratio and information ratio approximately 20% higher than portfolios with a top-down approach.

⁶In this paper, both portfolio construction techniques are compared using the value, momentum, quality and low-volatility factors over the period 1993-2015. Their results indicate that the bottom-up approach had a 20% higher information ratio than the top-down approach.

⁷In this paper, portfolios of long positions are compared using both portfolio construction techniques and applies them to a universe of MSCI World Index benchmark stocks over the period February 1993 to December 2015, using value and momentum as factors. The results of their work show that for all levels of tracking error, portfolios constructed with a bottom-up approach have higher information ratios than do top-down portfolios when both types of portfolios are matched on tracking error. Only for very low levels of tracking error (below 1%), the performance of both approaches would be similar.

⁸In this research, both approaches in portfolios of long positions are compared and applied to all NYSE, AMEX, and NASDAQ stocks during the period June 1963 to December 2016. The portfolios are constructed using 26 combinations of

portfolios constructed with a bottom-up approach do not generate statistically significantly higher Sharpe ratios than top-down portfolios do. Proponents of the top-down approach argue that it is more transparent and flexible and provides more information on the contribution to portfolio performance. Furthermore, Amenc et al. (2018) indicate that bottom-up approaches result in highly concentrated portfolios and that, owing to possible overfitting and multiple testing biases, the back tests of the bottom-up approach might be overestimated. However, Ghayur et al. (2018) stress that supporters of the bottom-up approach argue that this technique is superior to the top-down technique because it excludes stocks that have favourable exposure to one factor but could have unfavourable exposure to one or several of the other risk factors, which could neutralize the final risk premium. However, Ghayur et al. (2018) and Blitz & Vidojevic (2019) argue that the papers claiming superiority of the bottom-up approach do not make adequate comparisons because they do not compare (average) exposure matched portfolios (instead, the matching is done by employing a risk metric such as tracking error or volatility); if they do both types of approaches would offer similar performance.

Blitz & Vidojevic (2019) compared both portfolio construction approaches and an intermediate approach (enhanced single portfolio) using stocks traded on the NYSE, AMEX and NASDAQ exchanges for the period 1963-2017. They considered the factors of size, value, profitability, investment and momentum. The intermediate approach is the same as the top-down approach, but stocks with a given number of unfavourable characteristics with respect to other risk factors are removed from the individual factor portfolios. The authors conclude that the top-down approach is not optimal for obtaining factor exposures although they attribute this result to the fact that bottom-up portfolios are more concentrated, and therefore more effective at capturing larger factor exposures, outperforming top-down portfolios. However, using the intermediate approach, they conclude that when portfolios are matched on factor exposures, the construction technique applied is not relevant. Recently, Kim (2023) extended the approach of Blitz & Vidojevic (2019) to include factors such as size, value, momentum, profitability and low beta and US market data over the period 1970-2020, and conclude that the bottom-up approach is superior to the top-down and the intermediate approaches.

In addition, interest in the use of the investing factor is currently growing, although the results generate new unknowns instead of shedding light on the use of these financial management strategies. The following are worth mentioning:

- There is no consensus on whether to use a single reference factor or, on the contrary, implement a multi-factor strategy. Thus, DeMiguel et al. (2009) argue that a multifactor strategy using an equal-weighted scheme of factors is adequate and difficult to beat because it allows a balanced and diversified exposure

value, profitability, investment, momentum, and low-volatility factors. They conclude that the bottom-up approach achieves better performance than the top-down approach. However, when applying a multiple hypothesis framework they were unable to reject the null hypothesis that portfolios constructed with a bottom-up approach generate statistically significantly higher Sharpe ratios than those constructed with top-down approaches.

to different factors; on the other hand, Bender et al. (2010) argue that employing a combination of risk premia between different asset classes offers returns similar to a 60/40 equity bond benchmark but with 65% less volatility. Additionally, Alighanbari & Chia (2016) compared static multifactor strategies (equally weighted schemes of six factors: size, value, momentum, volatility, dividend and quality) and dynamic strategies, which, in this case, use fundamental data to weight each factor and find a superiority of dynamic strategies, as they offer higher returns, although at the cost of greater turnover and greater complexity; along these lines, Lewellen (2015) employs 10-year rolling Fama & MacBeth (1973) regressions based on 15 company characteristics to predict stock returns the following month; thus, the weight of a factor that ceased to be representative would gradually decrease to zero; however, tilting the weighting towards the stronger factors in the retrospective window and away from the weaker ones could lead to underperformance if the factors show a reversion to the mean in the long run. As a consequence of this time lag between the portfolio rebalancing period and the investment term, other authors (Arnott et al. (2016a), Arnott et al. (2016b) and Arnott et al. (2017)) propose dynamic approaches that include a factor-timing mechanism based on macroeconomic data, financials, sentiment, etc. In contrast, Asness et al. (2016) and Asness et al. (2017) emphasize the difficulty of factor timing and the perverse out-of-sample effects it could have when considering transaction costs. In summary, the question of whether factor timing or static long-run approximations should prevail as a criterion is unresolved and requires further research.

- Related to the previous question, two fundamental unresolved issues arise in the literature. On the one hand, is there any relationship between changes in risk factor exposures and the optimal rebalancing period of portfolios; on the other hand, is there a time structure of risk premia that depends on this optimal period? Flint & Vermaak (2023) show, for 5 factors in 12 markets, that over the last 20 years factor exposures decay over an approximate investment period of 3 years, although the period is very different depending on the factor, and moreover the effect is milder in single-factor indexed portfolios; however, when the investment periods are shorter, they observe significant dispersions in the results far from the expected distribution; more specifically, the value and quality factors show a slower decay, whereas investment and momentum are faster, but also show higher return margins. Thus, there is no clear consensus on how to choose the factor to reference in the portfolio and how to determine its decay, since the optimal rebalancing period differs from one portfolio to another.
- Another relevant issue is the diversifying capacity of factor investing strategies. Heinrich et al. (2021) show that a diversified factor portfolio can even show better performance than a portfolio with a concentration of factors, but this reduction in volatility is only possible if the portfolio only includes long positions in the assets. Rodrigues & Casalin (2022), in the Brazilian market, prove that diversification through factors offers higher returns than those obtained through geographic or sector diversification, for which the weighting of each factor within the portfolio is relevant. Bri  re & Szafarz (2019) compare multifactor portfolios (using size, value, profitability, investment and momentum factors) with sectors

investing in the US stock market over the period 1963-2014. The authors find that diversification with risk factors, when short sales are not restricted, outperforms sector diversification. Without the possibility of short sales, factor diversification performs better than sector diversification during bull periods but underperforms during bear periods. Thus, financial research has not reached a consensus on the risk diversification potential of factor investing strategies.

In this context, our initial hypothesis is that there are asymmetric risk premiums for the financial factors involved in asset valuation and, as a consequence, that factor investing type portfolio management can be designed to achieve superior performance to traditional passive management; we call this new type of management conditional factor investing.

The rest of the paper is organized as follows: Section 2 describes the methodology followed for the study. Section 3 describes the data used; Section 4 presents the results obtained and ends with the main conclusions drawn from the study.

2. Methodology

In our factor investing approach, we define a factor within a risk factor set as one that is involved in an asset pricing model and is therefore valued in the market through a portfolio composed of individual assets that best represents the financial characteristic with which the factor is identified. Thus, f is the continuous daily return of the factor and F is the total factor. On the other hand, we define r_i as the daily continuous return of asset i belonging to a market composed of N assets.

Then, we calculate the sensitivity of any asset- i to a factor- f through the corresponding β as:

$$\beta_{f,i,t-H} = \frac{\text{cov}(r_{i,t-H,1}, f_{t-H,1})}{\text{var}(f_{t-H})} \quad (1)$$

where cov is covariance, var is variance, $t-H$ is the calculation date of beta and J^9 is the sample size with which the beta is calculated (i.e., daily data are used from $t-H-J$ to $t-H$). Note that subindex 1 shows the daily frequency of the data.

We also define the expected excess return of an asset for an investment period H as the sum of the daily continuous returns for the period H (from $t-H$ to t) minus the risk-free rate in continuous compounding for the term H at instant t :

$$Er_{i,t,H} = \sum_{h=1}^H r_{i,t-h} - Rf_{t,H} \quad (2)$$

⁹It is common for individual assets, which do not trade every day, to start trading throughout a sample or to stop trading; to alleviate the liquidity problem this causes, as is usual in the literature, to calculate the beta of an asset, it must be traded for at least 50% of the sample total of each estimate (J).

Then, applying the asset pricing model of Fama & MacBeth (1973), we define the following model for any investment period H and at date t as:

$$Er_{i,t,H} = const + \sum_{f=1}^F \lambda_{f,t} \cdot \beta_{f,i,t-H} + u_{i,t,H} \quad (3)$$

where $\lambda_{f,t}$ is the risk premium associated with the factor- f at moment t . Note that the estimation is in the cross-section, so that we obtain as many estimates as the calculation instant, that is, so many periods of length H within the time interval from t to T .

Then, for a given confidence level (α) and at any instant t , the conditional factor investing strategy is realized for a factor- f such that: $tprob(\lambda_{f,t}) \leq \alpha$, where $tprob$ is the probability of a t-student distribution of the t-value of the corresponding risk premium, estimated as usual, that is, the quotient between the value of the parameter and its standard error. Note that this conditional strategy differs from other scoring strategies in that it makes the investment decision not on the basis of a ranking of previous normalized factor returns but on the basis of its statistical significance in predicting asset returns. The difference is that in the first case neither the predictive ability of the factor nor its lagged correlation with individual assets is considered.

Now, if the risk premium of the factor- f is statistically significant and positive, we define the conditional strategy as investing in the set of individual assets whose betas with respect to the factor- f at time t^{10} are higher than or equal to the P th percentile of the total calculated betas (N assets); on the other hand, if the premium is significant but negative, then positions are taken in those individual assets whose betas are lower than or equal to the $(100 - P)$ th percentile.

Thus, $w_{f,i,t}$ is the participation of asset- i in the conditional strategy of factor- f at time t , which only takes value 1 (in all other cases it is 0), if the following conditions are fulfilled and in the same order:

1. $tprob(\lambda_{f,t}) \leq \alpha$
2. According to the premium sign:
 - If $\lambda_{f,t} > 0$ then, $\beta_{f,t}^{Pth} \leq \beta_{f,i,t}$
 - Else $\lambda_{f,t} < 0$ then, $\beta_{f,i,t} \leq \beta_{f,t}^{(100-P)th}$

Note that by reducing the number of individual assets that can participate in the conditional strategy, depending on the required beta percentile, two objectives are achieved: on the one hand, transaction and portfolio liquidation costs are reduced, and on the other hand, the effect of the risk factor is amplified (positive premium) or mitigated (negative premium).

Then, the return obtained by the conditional factor- f investing strategy (cr_f) is calculated as:

$$cr_{f,t+H} = \frac{\sum_{i=1}^N w_{f,i,t} \cdot r_{i,t+H}}{\sum_{i=1}^N w_{f,i,t}} \quad (4)$$

¹⁰Note that to estimate the risk premiums the betas used are calculated in $t - H$, whereas to invest they are calculated in t . This is not illogical, since to estimate the premiums it is necessary to know the returns of the assets from $t - H$ to t , therefore, if the information is available, it is appropriate to use it to make the investment decision up to the $t + H$ instant.

Finally, the combined conditional factor investing strategy would be an equal-weighted portfolio of all factors whose risk premiums were statistically significant. Then $\theta_{f,t}$ takes value of 1 if the risk premium of factor- f was significant, and 0 otherwise. Thus the performance of the conditional combined conditional equal-weighted strategy CCe is:

$$CCe_{t+H} = \frac{\sum_{f=1}^F \theta_{f,t} \cdot cr_{f,t+H}}{\sum_{f=1}^F \theta_{f,t}} = \frac{\sum_{f=1}^F \theta_{f,t} \cdot \left(\frac{\sum_{i=1}^N w_{f,i,t} \cdot r_{i,t+H}}{\sum_{i=1}^N w_{f,i,t}} \right)}{\sum_{f=1}^F \theta_{f,t}} \quad (5)$$

3. Data

Our sample is composed of the daily rolling returns of 3285 stocks from the US market and 1257 stocks from the developed country market excluding the US market. The sample period spans from January 1, 2002 to June 30, 2023. The data were obtained from the Thomson Reuter's Refinitiv database, and all prices are in US dollars.

The developed ex-US markets sample consists of the following countries and assets, the latter included in the corresponding market index: Australia (181), Austria (19), Belgium (19), Canada (205), Denmark (19), Finland (25), France (37), Germany (40), Hong Kong (80), Ireland (17), Italy (39), Japan (225), Netherlands (21), Norway (64), New Zealand (44), Portugal (15), Singapore (30), Spain (35), Sweden (26), Switzerland (19) and the United Kingdom (97).

In the US market, 350 shares are traded on the New York Mercantile Exchange and 2935 shares are traded on the NASDAQ.

For the same time period and both markets, the daily data were downloaded from French's data web¹¹ the daily data for the risk-free rate (Rf) to calculate the excess of daily return over the risk-free rate of each asset, and the daily returns of factor size (SMB), growth (HML), conservatism vs aggressive investment (CMA), momentum (WML), market portfolio (Mkt) and robust minus weak operating profitability (RMW).

Finally and for both markets, the daily returns of the low market beta (BAB) and quality minus junk (QMJ) factors were downloaded from the AQR Capital Management web¹²

4. Results

First, we analyse how the factors behave with respect to the market factor, since the latter is the fundamental reference factor in asset pricing (CAPM). Table-1 shows the percentages of increases (positive returns) and decreases (negative returns) in each factor with respect to what happens with the market factor. All the results are expressed as continuous compounding returns for various investment terms (from 112 months).

[Table 1 about here.]

¹¹mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

¹²aqr.com/Insights/Datasets.

From Table-1, we observe the following:

- The up and down events are very similar for the overlapping and nonoverlapping estimates.
- As the time horizon increases, the number of up events increases, although in a much smaller proportion, than the number of down events.
- In the US market, the *Mkt* factor has the highest proportion of up events at all horizons, followed closely by the *BAB* factor. However, when *Mkt* is low, the *QMJ* factor has the highest percentage of up events, although this difference with respect to the other factors narrows as the time horizon increases.
- In development ex-US, this is not the same; there are several factors with more up events and fewer down events than *Mkt* (*QMJ*, *BAB*, *WML* and *RMW*).

Now, we analyse the mean return of each factor and investment period. Table-2 shows the results.

[Table 2 about here.]

From the results of Table-2, we observe that the average returns are very similar for both methods of calculation (overlapping and nonoverlapping). We also find that as the investment time horizon increases, the returns increase in a smaller proportion so that short time horizons seem to be more profitable. However, if we differentiate between the average return when the factor increases and when it decreases, we find that for the US market, there are certain factors whose ratio between the average return increases and the absolute value decreases is greater than 1 (*RMW*, *CMA* and especially *QMJ*); whereas for the development ex-US, the factors with this characteristic change (*QMJ*, *WML*, *HML* and especially *BAB*).

As some of the empirical works reviewed above show, it is possible that higher returns over shorter time horizons simply represent remuneration adapted to higher risk (timing vs. volatility). To test this, we estimate the Sharpe ratio (average return divided by standard deviation) for each factor and investment period. The results are shown in Table-3.

[Table 3 about here.]

From Table-3 shows that the Sharpe ratios are similar for both forms of calculation (overlapping and nonoverlapping). On the other hand, since the average returns do not consider the risk assumed to obtain this return, the Sharpe ratios show the average return obtained per unit of risk (measured by the standard deviation), and in this respect, we observe that while *Mkt* is one of the factors that remunerates more, especially in the short term, on the other hand, as the term increases, there are other factors that present higher remuneration per unit of risk; for example, *Mkt* upside in the US market for 6 months is 1.52, whereas *WML up + up* is 1.81; similarly, if we compare (quotient) the Sharpe ratio up with the down (in absolute value), we find that there are factors that show a value even higher than 2 (*WML* US for 6 months).

325 In summary, there are factors that show more up situations than the market (and less down), others
show a considerable percentage of up events when the market is down. In addition, there are factors that
show higher average returns than the market does, and these returns are proportionally higher in the short
term. Finally, per unit of risk (Sharpe ratio), the market is not the most remunerative factor in all cases.
Thus, factor-based portfolio management is justified, since it can allow better results than can the market
330 in downmarket situations and assume lower risks; the question is how to select the factor to be replicated in
asset management.

Table-4 below shows a summary of the betas calculated for each factor.

[Table 4 about here.]

For the results of the beta calculations in Table-4, we observe that as the sample size increases (from 12
335 months to 60 months, in the daily data), the extreme values decrease. This may be because by requiring
at least 50% contribution during each calculation period, some companies are excluded because of their low
representativeness (they quote less than half of each period), and logically, 120 days (12 months) versus
625 days (60 months), approximately, is a more demanding requirement. On the other hand, increasing the
number of observations to calculate the betas also results in greater convergence towards central values. If we
340 look at the calculations with 60 months, we again observe that the factors with more extreme betas (higher
and lower) are not the same for each of the markets considered (US and development ex-US markets).

From the calculated betas we estimate the risk premiums by applying the method of Fama & MacBeth
(1973). The results can be seen in Table-5 and Table-6.

[Table 5 about here.]

345 [Table 6 about here.]

From the results of the Table-5 and Table-6, we observe that the maximum number of companies partic-
ipating in each cross-section decreases as the sample size for calculating the betas increases; additionally, we
find that the explanatory power of the factors increases with the investment term and with the sample size
of the betas. Furthermore, with the exception of investments at 1, 3 and 6 months in the US market with
350 betas of 60 months, the constant is the parameter that is most often significant in cross-section. Among the
factors, a priori, it is to be expected that the *Mkt* factor (market portfolio) would be the most significant,
but this only occurs with 12-month betas and for an expected return of 3 months in the US market and for
betas of 60 months and an expected return of 12 months in the development ex-US market. Therefore, the
conditional factor investing strategy is fully justified.

355 Additionally, we observe that in the US market, for both 12-month and 60-month betas, up to 12-month
expected returns, only the *QMJ* factor is significant, for the longest return (12 months), as is *Mkt*, and for
12-month betas *CMA* is added. This coincides with the fact that *QMJ* is the factor with the most times of
statistical significance in the cross-section for those expected returns, although it is not the only one, nor does
it necessarily have to be the best performer. In contrast, in the development ex-US market, the significant

factor for 12-month betas for expected returns from 1 to 6 months, both inclusive, is *BAB*, whereas for 60-month betas it is *HML*; for the 12-month expected return *HML* and *QMJ* are added, for betas of 12 and 60 months, respectively; finally, note that the most significant factors in this market are not those that were statistically significant the most times in the cross-section. In summary, the betas calculated with 12-month data present higher volatilities and extreme values that can influence the statistical significance of the factors. Furthermore, the factors that are significant to the Fama-McBeth model are not always the same for all investment time horizons, nor are they necessarily the ones that are most often significant in cross-section, nor are they guaranteed to be the ones that present the best investment results.

Table-7 shows the Sharpe ratio for the returns obtained from 1 to 12 months for portfolios formed by those stocks whose betas with respect to each factor are above or below the 95th and 5th percentiles in each month of rebalancing.

[Table 7 about here.]

From the results in Table-7, we can see that the factors do not behave in the same way in the US and development ex-US markets; rather, we look at only the *WML* factor in the betas calculated with 12 months of daily returns. This difference disappears with respect to the amount of historical data used to calculate the betas. For both markets there are more cases with positive Sharpe ratio when 12 months of daily data are used. Notably, there is an increasing temporal evolution (temporal structure) of the Sharpe ratio; for example, while at an investment term of 1 month the portfolio-95% *WML* in development ex-US and betas 12 months has a value of 0.076; on the other hand for an investment with 12 months to maturity its value is 0.237; thus, the growth of the risk remuneration is more than proportional with respect to the investment term.

In summary, the investing factor requires differentiating by market and factor to set the investment term, and in some cases it is more profitable to invest in assets with lower betas with respect to the factor (5th percentile) and in others with higher betas (95th percentile); in particular, for the US market, portfolios with lower betas are preferable in the case of the *SMB* factor and higher betas for the remaining factors with positive returns (*HML*, *QMJ* and *RMW*). Regarding the development ex-US market, assets with higher betas for the *BAB* and *QMJ* factors and lower betas for the remaining factors with positive returns (*CMA*, *RMW* and *Mkt*) are preferable. Interestingly, the *Mkt* factor shows no risk remuneration in the US market. Finally, it seems more relevant to use more recent daily historical information (12 months) to calculate the betas.

Next, and only using the betas calculated with 12 months of daily data, we perform an experiment that consists of investing in the assets with extreme betas with respect to each factor in a month t and a term of $t + H$ (where H ranges from 1 month to 12 months), but only if the risk premium associated with the factor was statistically significant at 5% at least at instant t . Additionally, this strategy is complemented with the investment in the higher extreme betas if the significant premium was positive and in the lower extremes if it was negative. This second strategy combines the predictive capacity of each factor (first of the strategies

proposed) with the sign of its predictive influence, i.e., if the risk premium was positive (negative) it would imply that the factor has a positive influence on the expected return, so the higher (lower) the weight or betas with respect to the factor, the higher the expected return. The results can be seen in Table-8.

[Table 8 about here.]

When we analyse the results of Table8 for the first of the strategies, investing in factors with significant premiums, we observe that the highest remuneration per unit of risk occurs in the 6 and 12 month terms, with the exception of the *SMB* factor for the US market, which does so in the 3-month term. However, if we add to this conditionality the sign of the premium to take one extreme or the other of the corresponding beta distribution, we find that not only in some cases, there is a change in the optimal investment term (for example, CMA in the US market goes from 12 months to 1 month), but in all cases the remuneration per unit of risk in this second conditional strategy is much higher than in the first.

In short, factor investing requires estimating the statistical significance of the factors in an asset pricing model at the chosen investment term and then making a decision to place assets in those whose betas with respect to the significant factors are higher or lower depending on the sign of the risk premium (positive or negative, respectively). By way of example, and the basis of our results, an investor interested in the US market for a 12-month term should place his investment in assets with the lowest (5th percentile) beta with respect to the QMJ factor, but when its associated risk premium in the Fama-MacBeth model on expected excess return on the risk-free rate at 12 months, is significant and negative; thus obtaining a remuneration of 1.563 per unit of risk.

Now, we test whether the conditional factor investing strategy also shows a better return per unit of risk, but unlike the previous results, we will not assume a monthly rebalancing but rather a buy-and-hold investment for the entire investment period (nonoverlapping). Additionally, the conditional strategy will be to invest only if the risk premium associated with the factor is significant at least at the 5% confidence level, in which case, if the premium is positive, individual assets with betas equal to or above the 95th percentile are purchased, whereas if the premium is negative, the assets purchased will be those whose betas are equal to or below the 5th percentile. Therefore, unlike the previous results, not only is the investment nonoverlapping, but also the strategy with respect to each factor and investment term can be made on the upper or lower end of the distribution of betas, depending on the sign of the premium. The results can be seen in Table-9.

[Table 9 about here.]

From Table-9, we highlight that the strategy conditioned to the prior statistical significance of the risk premium always involves a lower number of investment operations with respect to the replication of the factor, i.e., for example, if we replicate the 12-month Mkt factor with betas calculated with 60 months of daily data, we will perform 15 rebalances (every 12 months nonoverlapping of the sample period), whereas if the strategy is conditional, only 8 rebalances of the portfolio will be performed (there will be 7 periods in

which the nonoverlapping of the risk premium will occur), we will perform 15 rebalances (every 12 months nonoverlapping of the sample period), whereas if the strategy is conditional only 8 portfolio rebalances are performed (there will be 7 periods in which the premium is not significant and will not be invested). Therefore, in terms of transaction costs for all factors, investment terms and betas calculation periods, the conditional strategy is less expensive than the factor replication is.

However, a lower transaction cost is not enough to test the efficiency of the conditional strategy versus factor replication; there should also be a higher remuneration per unit of risk.

For the US market, we again observe that as the investment term increases the results are more consistent with larger sample sizes for calculating betas. In particular we highlight the following:

- For investment terms of 1 month there is no factor that outperforms the conditional strategy with 12-month betas, whereas for 60-month betas we observe that *Mkt* and *QMJ* show higher remuneration per unit of risk.
- For 3-month investment terms and 12-month betas, the *BAB* factor and *Mkt* show higher remuneration per risk, whereas for 60-month betas, in addition to the above two, *QMJ* remains the same as for the previous term and *RMW* also outperforms the strategy.
- For the 6-month terms and 12-month betas, in addition to the *BAB* and *Mkt* of the previous term, *RMW* appears. In this context, the factors with the highest remuneration per unit of risk between 12 and 60 month betas also coincide.
- For the 12-month terms and both betas (12 and 60 months) the factors with the highest returns per unit of risk are *BAB*, *CMA*, *RMW* and *Mkt*.

In summary, with respect to the US market, we find that the conditional strategy is the most remunerative in the short term (1 month) and short databases (12 months); as the investment term increases, the *Mkt* and *BAB* factors are the most remunerative per unit of risk regardless of the sample size. In addition, we observe that there are factors whose remuneration increases significantly for longer terms (from 6 months onwards) such as *CMA* and *RMW*. Finally, although *Mkt* and *BAB* offer higher risk remunerations, we should not forget that their transaction costs are also higher as the number of operations to replicate them is at least 1.6 times greater.

For the development ex-USA market, we again find that, as the sample size for calculating betas increases, the conditional strategy obtains worse results, since with betas of 60 months of daily data we observe that *BAB*, *QMJ* and *WML* remunerate more per unit of risk regardless of the investment term, and in some longer terms *RMW* and *Mkt* do so as well. Undoubtedly, the geographic diversity of this sample with respect to the US market results in better results for factors such as the *QMJ* and *WML* show better results, whereas the *Mkt* factor has worse remuneration results by risk unit.

If we look at the 12-month calculation period, we obtain the same conclusions as for the US market, i.e., the *BAB* and *Mkt* factors remunerate more per unit of risk, and additionally, for some longer investment

periods, other factors such as *HML*, *QMJ* and *WML* are added; without forgetting again, replicating these factors that remunerate more entails higher transaction costs, since the number of operations to replicate them is at least 1.75 times greater than the factor's conditional strategy.

Since the conditional factor investing strategy shows a lower number of investments than continuously replicating the factor, with the consequent lower transaction costs, the question arises as to whether this savings is only a consequence of conditionality (statistical significance of the factors' risk premiums) or whether it is also the result of the fact that a limited number of assets participate in the total and, as a consequence, there are certain individual assets associated especially with each strategy. To check this, Table-10 shows for each factor and investment term the number of individual assets that participate at least once in the strategy and the individual asset that participates the most together with its percentage of participation.

[Table 10 about here.]

Table-10 shows that as we increase the investment term, the number of individual assets participating at least once in each strategy decreases; for example, a 1-month *BAB* in the US market with betas of 12 months of daily historical data has used 1865 assets out of the total of 3285, whereas a 12-month *BAB* has used only 595 assets. Moreover, when the strategy is performed with betas calculated with 60 months of daily historical data, the participation of individual assets is lower than when 12-month betas are used, probably because of the higher data volume requirement (at least 50% of the total sample); thus, the strategy based on smaller databases suffers from higher volatility in terms of individual assets, whereas larger databases do not participate in the potential positive returns of individual assets with little history.

Regarding the factors that require more individual assets for the conditional strategy, we can observe that in the US market it is *BAB* at 1 month (1865 assets or 57% of 3285 assets) for 12-month betas and *WML* at 1 month (986 assets or 30% of 3285 assets) with 60-month betas. The development ex-US market, on the other hand, shows that with 12-month betas it is the *WML* factor at 1 month (931 assets or 74% of 1257 assets) and for 60-month betas it is the *CMA* at 1 month (537 assets or 43% of 1257 assets).

As for the factors using fewer individual assets, we find that in the US market, for both 12-month and 60-month betas, it is 12-month *Mkt* (330 and 337 assets or 10% of 3285 assets), while for the development ex-US market it is the 12-month *RMW* factor (287 assets or 23% of 1257 assets) and 12-month *QMJ* (190 assets or 15% of 1257 assets), with 12-month and 60-month betas, respectively.

From the above results we observe that in some cases, for both markets, the *BAB*¹³ and *Mkt* factors remunerate more per unit of risk, but only in the case of buy-and-hold investment until the end of the

¹³Importantly Fama-French factors (*SMB*, *HML*, *CMA*, *WML*) are formed yearly (June 30). To be comparable, they must take that frequency since in certain cases they incorporate accounting information that is presented with that frequency; also, the *Mkt* factor is a value weight, therefore, like the previous ones, value weighting is performed annually. Similarly, the *QMJ* factor is also formed annually when accounting information is used, among other factors. On the other hand, the *BAB* factor is formed monthly, which, a priori, seems to give it a certain advantage in terms of the updated information it incorporates in its returns.

investment term (nonoverlapping). However, we cannot only compare the result of investing in a single factor with its corresponding conditional strategy, since the asset pricing model, on which the conditional strategy is based, is multifactorial; therefore, at each investment moment we do not take a position only with respect to one factor, but we invest in all the conditional strategies whose associated factors present a statistically significant risk premium (if it is positive we buy the assets with higher betas and if it is negative with lower betas, according to the chosen percentile, which in our case are 95th and 5th percentiles). Thus, Table-11 shows the Sharpe ratio for such a combined and conditional strategy, constructed as an equal-weighted investment in the conditional strategies of each factor with a significant premium. We include the two factors with the highest Sharpe ratios to facilitate comparisons.

[Table 11 about here.]

From the results in Table-11, we find that the conditional strategy is superior for the short terms (1-3 months) and mainly uses daily data for the last 12 months to calculate the betas. This is therefore clear evidence that the conditional strategy is subject to the predictive ability of the factors used, i.e., when the term of the expected excess return used in the Fama & MacBeth (1973) model increases, the long term predictive ability decreases and consequently, the results obtained from the factor investing strategy.

On the other hand, if we look at the investments with 12 months of daily data to calculate the betas, we observe that the US market combined strategy obtains better results than the development ex-US strategy does; in the former, as would be expected according to financial theory, the market factor (*Mkt*) is the most remunerative, whereas in the latter, the *BAB* factor presents better results, no doubt owing to the very different basket of countries that make up this market.

To conclude our empirical study, we compare the results graphically for the investment term (6 months) from which it is observed that the remuneration per unit of risk is higher for some factors than for the factor investing strategy. This comparison consists of a recurrent nonoverlapping investment with an initial capital of \$100 and reinvesting the result at maturity (every 6 months) until the end of the sample period. The strategies we analyse as follow: combined conditional equal-weighted (*CCE*), equal-weighted in all the factors of the model (*FE*) and only in the factor that pays more in the long run, i.e. *Mkt* and *BAB* for the *US* and development ex-US (*DEV*) markets, respectively. In this regard, let us recall that, while investing in the factors requires that at all maturities a sale of the assets acquired and a purchase of the assets for the next investment period is carried out, in the case of the conditional strategy this will only occur if the risk premium is significant; moreover, as we have seen above, the number of individual assets involved is lower; in other words, the transaction costs will be much lower for the conditional strategy; in contrast, if no investment is made in a period (no significant premium) the remuneration for that period would be 0%. The results are shown in Figure-1.

[Figure 1 about here.]

The large difference between the strategies is obvious, while the combined conditional factor investing strategy earns more than 185% per annum and 109% annually in the development ex-US and US markets,

respectively, the best single factor investing strategy is between 50% per annum (BAB_{DEV}) and 30% per annum (Mkt_{US}).

5. Conclusions

Within the financial literature there is no consensus on the factor investing strategy since there is no prior definition of what a factor is. In our work we have provided a clarification in this regard, we define factor investing as an investment strategy that replicates the behaviour of a significant risk factor in an asset pricing model. This is relevant, since, for example, if we define the book-to-market ratio (size) as a factor, investing in companies with a high ratio would not mean that we are replicating a pricing factor, since this variable (in this case SMB) should be represented by the profitability of a portfolio of assets selected according to this financial characteristic and be significant in an asset pricing model (prediction of excess return), since in this way, the associated risk premium would indicate the expected remuneration for this factor.

Following this reasoning, we define a conditional strategy consisting of investing in individual assets that replicate a certain factor only if the associated risk premium is significant. Furthermore, considering the sign of this premium, the investment is made in assets with betas above a high percentile if it is positive; in contrast, if it is negative, positions would be taken in those assets with premiums below a low percentile. In this way, not only is the expected effect of the risk premium amplified but also, as fewer individual assets are involved, transaction costs are reduced.

Finally, the combined strategy is an equal-weighted portfolio of those assets that should be invested in following the criteria defined above for each of the factors.

Based on this novel proposal, an empirical study is conducted for the US (3285 assets) and development ex-US (1257 assets) equity markets from January 1, 2002 through June 30, 2023, using betas calculated from 12-month and 60-month daily compound returns.

The results show that for short investment terms (1-3 months) and from the betas calculated with more current data (12 months of daily data) the remuneration per unit of risk of the conditional strategy outperforms the replicated factors. For longer terms (6-12 months) the market portfolio factor (Mkt) in the US and the BAB factor in the case of development ex-US show greater remuneration per unit of risk. However, we have found that this remuneration does not correspond to the cumulative realized result of a periodic reinvestment, as the combined conditional factor investing strategy achieves much higher capital gains than these factors do.

In summary, this paper aims to shed light on the factor investing strategy, which can help investors and portfolio managers. In addition, it opens a line of research in which different questions remain to be solved, such as investment in terms of less than 1 month, the dynamic design of the strategy on the basis of the prediction of significant risk premia or the selection of the maximum and minimum percentiles of the betas to replicate the behaviour of the factors in an amplified way.

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Figure 1: Comparison of the cumulative results of investing \$100 over 6 months using the combined conditional equal-weighted factor investing, factor equal-weighted factor investing and factor with higher remuneration per unit of risk strategies

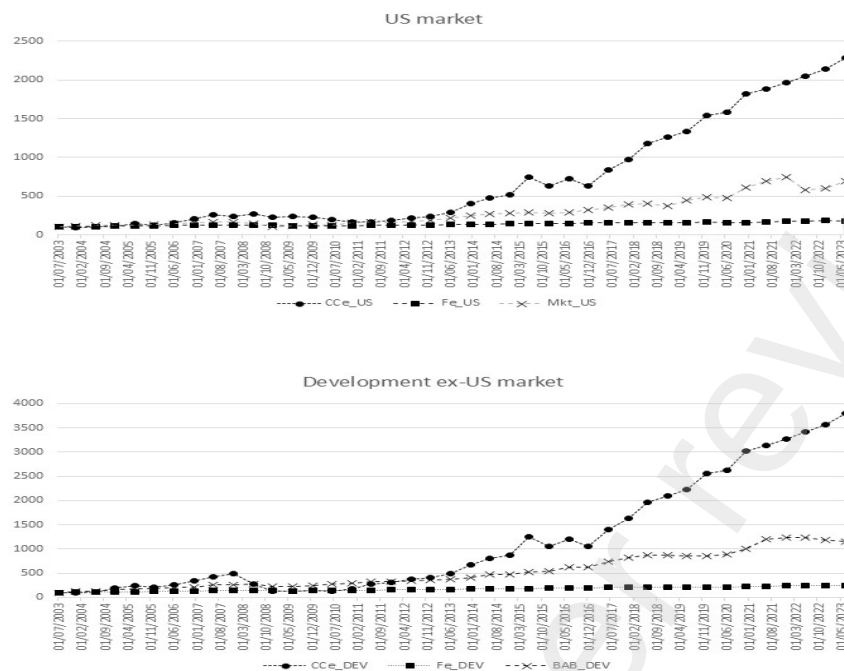


Table 1: Percentage of factor events and market factor events (expressed in %)

<i>US</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
<i>1m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	38.37	30.62	32.17	32.56	32.95	40.31	27.13	65.50	38.37	30.62	32.17	32.56	32.95	40.31	27.13	65.50
<i>u + d</i>	15.12	14.73	25.19	18.60	23.26	23.64	27.91	0.00	15.12	14.73	25.19	18.60	23.26	23.64	27.91	0.00
<i>d + u</i>	27.13	34.88	33.33	32.95	32.56	25.19	38.37	0.00	27.13	34.88	33.33	32.95	32.56	25.19	38.37	0.00
<i>d + d</i>	19.38	19.77	9.30	15.89	11.24	10.85	6.59	34.50	19.38	19.77	9.30	15.89	11.24	10.85	6.59	34.50
<i>3m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	41.02	36.72	33.20	32.81	38.28	49.61	30.08	69.92	44.19	41.86	33.72	37.21	40.70	53.49	29.07	72.09
<i>u + d</i>	7.81	14.06	23.83	15.63	19.92	19.53	26.95	0.00	9.30	12.79	22.09	12.79	22.09	18.60	26.74	0.00
<i>d + u</i>	28.91	33.20	36.72	37.11	31.64	20.31	39.84	0.00	27.91	30.23	38.37	34.88	31.40	18.60	43.02	0.00
<i>d + d</i>	22.27	16.02	6.25	14.45	10.16	10.55	3.13	30.08	18.60	15.12	5.81	15.12	5.81	9.30	1.16	27.91
<i>6m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	43.08	41.50	41.50	39.92	38.74	62.85	35.57	73.91	46.51	37.21	39.53	37.21	39.53	58.14	32.56	72.09
<i>u + d</i>	8.70	11.46	21.74	13.04	18.97	12.65	22.92	0.00	6.98	11.63	23.26	13.95	20.93	13.95	23.26	0.00
<i>d + u</i>	30.83	32.41	32.41	33.99	35.18	11.07	38.34	0.00	25.58	34.88	32.56	34.88	32.56	13.95	39.53	0.00
<i>d + d</i>	17.39	14.62	4.35	13.04	7.11	13.44	3.16	26.09	20.93	16.28	4.65	13.95	6.98	13.95	4.65	27.91
<i>12m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	45.34	41.30	51.01	36.44	43.72	73.68	38.46	79.76	38.10	38.10	57.14	33.33	38.10	76.19	42.86	80.95
<i>u + d</i>	8.10	7.29	18.22	13.36	16.60	9.31	19.03	0.00	9.52	14.29	14.29	19.05	19.05	14.29	19.05	0.00
<i>d + u</i>	34.41	38.46	28.74	43.32	36.03	6.07	41.30	0.00	42.86	42.86	23.81	47.62	42.86	4.76	38.10	0.00
<i>d + d</i>	12.15	12.96	2.02	6.88	3.64	10.93	1.21	20.24	9.52	4.76	4.76	0.00	0.00	4.76	0.00	19.05
<i>DEV.</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
<i>1m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	26.74	33.72	34.11	26.36	34.50	41.47	24.42	58.91	26.74	33.72	34.11	26.36	34.50	41.47	24.42	58.91
<i>u + d</i>	25.58	21.71	31.40	25.97	25.97	28.68	34.88	0.00	25.58	21.71	31.40	25.97	25.97	28.68	34.88	0.00
<i>d + u</i>	32.17	25.19	24.81	32.56	24.42	17.44	34.50	0.00	32.17	25.19	24.81	32.56	24.42	17.44	34.50	0.00
<i>d + d</i>	15.50	19.38	9.69	15.12	15.12	12.40	6.20	41.09	15.50	19.38	9.69	15.12	15.12	12.40	6.20	41.09
<i>3m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	34.77	42.19	43.36	33.20	36.33	48.83	33.98	63.28	34.88	39.53	47.67	30.23	38.37	51.16	34.88	65.12
<i>u + d</i>	15.23	18.36	27.34	23.83	25.78	22.27	34.77	0.00	11.63	18.60	27.91	24.42	25.58	24.42	32.56	0.00
<i>d + u</i>	28.52	21.09	19.92	30.08	26.95	14.45	29.30	0.00	30.23	25.58	17.44	34.88	26.74	13.95	30.23	0.00
<i>d + d</i>	21.48	18.36	9.38	12.89	10.94	14.45	1.95	36.72	23.26	16.28	6.98	10.47	9.30	10.47	2.33	34.88
<i>6m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	38.74	49.41	46.64	38.74	41.11	58.50	42.29	67.19	41.86	51.16	48.84	44.19	44.19	58.14	41.86	69.77
<i>u + d</i>	10.28	14.23	30.04	19.76	26.09	20.55	31.62	0.00	9.30	11.63	30.23	16.28	27.91	27.91	30.23	0.00
<i>d + u</i>	28.46	17.79	20.55	28.46	26.09	8.70	24.90	0.00	27.91	18.60	20.93	25.58	25.58	11.63	27.91	0.00
<i>d + d</i>	22.53	18.58	2.77	13.04	6.72	12.25	1.19	32.81	20.93	18.60	0.00	13.95	2.33	2.33	0.00	30.23
<i>12m.</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>u + u</i>	40.89	49.39	46.96	39.27	46.15	61.54	44.13	63.97	42.86	52.38	52.38	38.10	52.38	61.90	52.38	66.67
<i>u + d</i>	11.34	13.77	31.98	20.65	31.58	23.48	36.03	0.00	9.52	14.29	33.33	19.05	28.57	23.81	33.33	0.00
<i>d + u</i>	23.08	14.57	17.00	24.70	17.81	2.43	19.84	0.00	23.81	14.29	14.29	28.57	14.29	4.76	14.29	0.00
<i>d + d</i>	24.70	22.27	4.05	15.38	4.45	12.55	0.00	36.03	23.81	19.05	0.00	14.29	4.76	9.52	0.00	33.33

Note: *US* represents the factors for the US market; *DEV.* represents the factors for the developed countries market excluding the U.S.; *u + u* means the

percentage of times the factor goes up and the market goes up; *u + d* shows the percentage of cases where the factor goes up and the market goes down;

d + u is the percentage of cases where the factor goes down and the market goes up and; *d + d* is the percentage of cases where the factor goes down and

the market goes down at the same time. The returns have been estimated in continuous compounding for 1, 3, 6 and 12 months (*m.*). *Overlapping* means

that the estimates start every month, while *Nonoverlapping* means that until one monthly investment period (from 1 to 12 months) ends, the next one

does not start.

Table 2: Annualized mean return of factor events and market factor events (expressed in %)

<i>US</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
1m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	24.68	24.56	15.36	15.64	26.12	23.63	16.59	38.71	24.68	24.56	15.36	15.64	26.12	23.63	16.59	38.71
$u + d$	19.28	35.26	25.98	22.81	44.63	33.61	34.20	<i>n.a.</i>	19.28	35.26	25.98	22.81	44.63	33.61	34.20	<i>n.a.</i>
$d + u$	-20.42	-21.24	-17.84	-15.36	-48.10	-26.31	-23.85	<i>n.a.</i>	-20.42	-21.24	-17.84	-15.36	-48.10	-26.31	-23.85	<i>n.a.</i>
$d + d$	-30.25	-30.96	-18.32	-17.07	-29.00	-32.24	-18.63	-48.85	-30.25	-30.96	-18.32	-17.07	-29.00	-32.24	-18.63	-48.85
3m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	13.91	14.46	10.97	11.09	14.57	16.42	12.63	24.23	13.25	13.43	12.07	12.10	12.07	15.85	14.41	25.29
$u + d$	16.82	22.00	17.69	17.12	28.29	23.20	22.43	<i>n.a.</i>	11.27	23.44	18.58	15.94	30.08	24.31	22.52	<i>n.a.</i>
$d + u$	-9.94	-16.83	-9.88	-8.39	-30.82	-14.81	-14.68	<i>n.a.</i>	-10.07	-17.91	-9.18	-8.88	-32.40	-14.96	-14.86	<i>n.a.</i>
$d + d$	-14.57	-22.81	-10.64	-11.29	-15.52	-23.28	-12.61	-28.29	-16.71	-26.62	-14.01	-12.44	-22.76	-30.93	-13.03	-34.85
6m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	10.26	10.29	6.24	7.37	11.24	12.03	8.13	18.95	8.19	11.31	6.99	8.17	6.91	13.14	8.12	19.52
$u + d$	6.93	15.00	14.48	12.98	20.77	21.09	21.47	<i>n.a.</i>	12.35	19.41	13.40	16.32	25.02	25.78	21.33	<i>n.a.</i>
$d + u$	-7.28	-12.86	-6.87	-6.31	-22.09	-6.80	-10.62	<i>n.a.</i>	-5.43	-12.63	-5.94	-8.17	-22.79	-8.47	-8.94	<i>n.a.</i>
$d + d$	-10.94	-17.46	-4.21	-7.79	-13.13	-19.04	-6.88	-21.29	-11.00	-17.68	-2.35	-6.50	-6.95	-19.45	-8.61	-19.95
12m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	7.31	8.55	4.63	6.15	8.18	10.80	7.42	15.49	7.99	9.46	5.01	7.71	11.15	10.97	8.39	15.56
$u + d$	3.10	15.33	11.23	9.45	15.09	11.97	17.60	<i>n.a.</i>	2.45	13.42	17.24	11.62	15.98	13.87	19.70	<i>n.a.</i>
$d + u$	-5.09	-10.87	-4.83	-4.39	-14.82	-6.54	-7.15	<i>n.a.</i>	-4.18	-11.13	-5.83	-5.34	-15.17	-4.85	-8.56	<i>n.a.</i>
$d + d$	-7.33	-8.46	-2.22	-4.20	-37.88	-18.63	-1.82	-16.76	-5.17	-11.50	-2.01	<i>n.a.</i>	<i>n.a.</i>	-41.77	<i>n.a.</i>	-24.32
<i>DEV.</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
1m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	13.93	17.88	22.34	9.95	9.30	28.06	12.30	42.24	13.93	17.88	22.34	9.95	9.30	28.06	12.30	42.24
$u + d$	17.26	21.71	31.07	18.37	15.92	25.84	25.46	<i>n.a.</i>	17.26	21.71	31.07	18.37	15.92	25.84	25.46	<i>n.a.</i>
$d + u$	-17.64	-14.40	-37.10	-11.18	-10.75	-23.88	-16.72	<i>n.a.</i>	-17.64	-14.40	-37.10	-11.18	-10.75	-23.88	-16.72	<i>n.a.</i>
$d + d$	-18.42	-20.47	-20.12	-10.77	-10.78	-14.80	-8.51	-45.47	-18.42	-20.47	-20.12	-10.77	-10.78	-14.80	-8.51	-45.47
3m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	8.69	10.54	14.35	5.75	7.40	24.04	9.25	26.26	10.72	11.12	14.19	6.46	8.06	22.70	9.81	26.23
$u + d$	10.56	16.60	20.11	13.54	9.40	17.86	16.95	<i>n.a.</i>	9.15	16.14	18.35	13.18	7.78	15.59	17.40	<i>n.a.</i>
$d + u$	-9.29	-9.50	-23.03	-6.82	-5.09	-7.99	-11.43	<i>n.a.</i>	-10.34	-8.64	-25.66	-6.20	-5.04	-5.86	-10.89	<i>n.a.</i>
$d + d$	-11.03	-13.15	-11.16	-8.01	-6.69	-10.13	-7.05	-28.23	-8.89	-12.54	-17.05	-8.43	-6.96	-14.89	-8.68	-31.18
6m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	6.40	8.23	11.75	4.13	4.88	19.60	6.92	20.13	6.42	8.90	11.53	4.59	5.62	20.09	6.84	19.76
$u + d$	7.97	15.62	14.54	13.80	7.58	14.91	15.38	<i>n.a.</i>	7.78	15.44	13.40	15.79	5.92	9.52	15.53	<i>n.a.</i>
$d + u$	-6.15	-7.63	-16.93	-5.77	-3.09	-5.03	-9.07	<i>n.a.</i>	-6.72	-5.43	-16.54	-6.13	-3.08	-4.42	-7.06	<i>n.a.</i>
$d + d$	-8.38	-10.32	-12.74	-5.91	-3.81	-8.77	-3.20	-22.63	-9.20	-11.77	<i>n.a.</i>	-6.46	-11.30	-33.37	<i>n.a.</i>	-25.08
12m.	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
$u + u$	4.52	7.06	10.61	3.25	3.29	17.77	5.23	17.10	5.53	6.15	10.81	3.26	3.38	16.91	4.37	16.66
$u + d$	6.11	12.66	10.55	10.36	5.76	12.59	11.45	<i>n.a.</i>	5.84	17.45	10.71	15.88	6.82	16.50	13.46	<i>n.a.</i>
$d + u$	-5.03	-6.50	-12.11	-4.72	-2.26	-3.70	-5.81	<i>n.a.</i>	-5.18	-11.87	-20.37	-6.48	-2.18	-2.61	-8.36	<i>n.a.</i>
$d + d$	-6.67	-7.08	-12.16	-4.00	-2.22	-7.58	<i>n.a.</i>	-13.21	-7.15	-4.03	<i>n.a.</i>	-2.13	-5.34	-8.74	<i>n.a.</i>	-15.81

Note: *US* represents the factors for the U.S. market; *DEV.* represents the factors for the developed countries market excluding the U.S.; $u + u$ means the mean return of times the factor goes up and the market goes up; $u + d$ shows the mean return when the factor goes up and the market goes down; $d + u$ is the mean return when the factor goes down and the market goes up and; $d + d$ is the mean return of cases where the factor goes down and the market goes down at the same time. The returns have been estimated as annualized continuous compounding for 1, 3, 6 and 12 months (*m.*). *Overlapping* means that the estimates start every month, while *Nonoverlapping* means that until one monthly investment period (from 1 to 12 months) ends, the next one does not start. *n.a.* means not available.

Table 3: Sharpe ratio by factor events and market factor events

<i>US</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
<i>1m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.41	1.05	1.07	1.10	1.22	1.17	1.06	1.31	1.41	1.05	1.07	1.10	1.22	1.17	1.06	1.31
<i>u + d</i>	0.98	1.15	1.20	1.10	1.24	0.95	1.56	n.a.	0.98	1.15	1.20	1.10	1.24	0.95	1.56	n.a.
<i>d + u</i>	-1.50	-1.02	-1.09	-1.11	-0.82	-0.84	-0.96	n.a.	-1.50	-1.02	-1.09	-1.11	-0.82	-0.84	-0.96	n.a.
<i>d + d</i>	-1.55	-0.87	-1.23	-1.67	-0.98	-1.01	-0.95	-1.16	-1.55	-0.87	-1.23	-1.67	-0.98	-1.01	-0.95	-1.16
<i>3m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.10	1.04	1.06	1.36	1.19	1.29	1.12	1.50	1.13	0.83	0.94	1.18	1.25	1.26	1.20	1.44
<i>u + d</i>	1.72	1.28	1.45	1.01	1.47	0.97	1.42	n.a.	1.51	1.36	1.66	0.99	1.26	0.98	1.36	n.a.
<i>d + u</i>	-1.29	-1.19	-1.15	-1.37	-0.79	-1.00	-1.05	n.a.	-1.31	-1.50	-1.18	-1.30	-0.87	-1.10	-1.11	n.a.
<i>d + d</i>	-1.44	-0.97	-1.68	-1.44	-1.20	-1.01	-1.74	-1.01	-1.58	-1.01	-1.70	-1.81	-1.03	-1.29	n.a.	-1.18
<i>6m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.12	1.12	1.10	1.40	1.81	1.39	1.20	1.52	1.23	1.33	1.01	2.08	1.47	1.44	0.89	1.68
<i>u + d</i>	1.32	0.96	1.34	0.90	1.43	0.93	1.75	n.a.	1.69	1.52	1.08	1.12	1.53	0.88	2.21	n.a.
<i>d + u</i>	-1.34	-1.15	-1.08	-1.39	-0.79	-1.06	-1.10	n.a.	-1.22	-1.17	-1.18	-1.32	-0.87	-1.01	-1.42	n.a.
<i>d + d</i>	-1.93	-0.98	-1.13	-1.69	-0.66	-1.23	-1.57	-0.92	-1.50	-0.84	-2.14	-1.23	-2.00	-0.86	-2.84	-1.03
<i>12m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.18	1.56	1.04	1.64	1.59	1.48	1.47	1.72	1.55	1.55	0.86	2.69	1.71	1.68	1.48	1.74
<i>u + d</i>	2.10	1.63	1.83	1.03	1.79	1.17	2.44	n.a.	2.39	1.52	2.04	1.20	3.54	0.80	3.20	n.a.
<i>d + u</i>	-1.23	-1.23	-0.94	-1.51	-0.83	-1.39	-1.12	n.a.	-2.10	-1.13	-1.19	-1.66	-0.73	n.a.	-1.46	n.a.
<i>d + d</i>	-1.81	-1.18	-5.06	-1.58	-1.53	-1.60	-2.44	-1.12	-3.01	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	-1.70
<i>DEV.</i>	<i>Overlapping</i>								<i>Nonoverlapping</i>							
<i>1m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.19	1.24	1.50	1.08	1.23	1.26	1.30	1.40	1.19	1.24	1.50	1.08	1.23	1.26	1.30	1.40
<i>u + d</i>	1.39	0.90	1.15	0.97	1.66	1.17	1.51	n.a.	1.39	0.90	1.15	0.97	1.66	1.17	1.51	n.a.
<i>d + u</i>	-1.33	-0.98	-0.95	-1.03	-1.30	-1.07	-0.95	n.a.	-1.33	-0.98	-0.95	-1.03	-1.30	-1.07	-0.95	n.a.
<i>d + d</i>	-1.42	-1.20	-1.46	-1.32	-0.99	-0.79	-1.34	-1.02	-1.42	-1.20	-1.46	-1.32	-0.99	-0.79	-1.34	-1.02
<i>3m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.27	1.31	1.48	1.02	1.44	1.56	1.45	1.41	1.36	1.19	1.41	0.90	1.42	1.39	1.48	1.34
<i>u + d</i>	1.63	1.05	1.21	0.82	1.51	1.20	1.44	n.a.	1.70	1.06	1.17	0.85	1.32	1.35	1.49	n.a.
<i>d + u</i>	-1.41	-1.21	-0.75	-0.80	-1.30	-0.84	-1.02	n.a.	-1.76	-0.95	-0.91	-0.80	-1.25	-0.93	-0.97	n.a.
<i>d + d</i>	-1.33	-1.04	-1.11	-1.07	-1.11	-0.91	-2.45	-0.89	-1.39	-1.13	-1.06	-1.03	-0.93	-1.23	-7.93	-0.99
<i>6m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.48	1.35	1.68	1.27	1.30	1.77	1.55	1.39	1.50	1.27	1.50	0.84	1.26	1.55	1.59	1.64
<i>u + d</i>	1.62	1.21	1.35	1.01	1.46	1.13	1.71	n.a.	1.51	1.02	2.04	0.87	1.23	1.36	1.84	n.a.
<i>d + u</i>	-1.27	-0.94	-0.72	-0.82	-0.97	-1.34	-0.99	n.a.	-1.59	-1.06	-0.71	-0.75	-0.81	-1.75	-1.12	n.a.
<i>d + d</i>	-1.62	-1.11	-0.65	-0.94	-1.06	-0.87	-0.73	-0.90	-1.64	-1.07	n.a.	-0.90	n.a.	n.a.	n.a.	-1.08
<i>12m.</i>	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>u + u</i>	1.37	1.94	1.90	1.70	1.43	2.41	1.53	1.75	2.21	1.75	2.01	1.58	1.44	2.32	1.48	1.83
<i>u + d</i>	1.58	1.62	1.63	1.33	1.80	1.44	2.02	n.a.	2.62	2.57	1.31	1.67	1.72	1.76	1.98	n.a.
<i>d + u</i>	-1.60	-0.95	-0.85	-0.97	-1.44	-1.73	-1.20	n.a.	-3.42	-1.61	-1.18	-0.96	-1.84	n.a.	-2.01	n.a.
<i>d + d</i>	-1.81	-1.27	-0.97	-1.01	-1.72	-1.13	n.a.	-0.86	-2.21	-1.31	n.a.	-1.24	n.a.	-1.96	n.a.	-1.04

Note: *US* represents the factors for the US market; *DEV.* represents the factors for the developed countries market excluding the U.S.; *u + u* means the mean return of times the factor goes up and the market goes up; *u + d* shows the mean return when the factor goes up and the market goes down; *d + u* is the mean return when the factor goes down and the market goes up and; *d + d* is the mean return of cases where the factor goes down and the market goes down at the same time. The returns have been estimated are annualized continuous compounding for 1, 3, 6 and 12 months (*m.*). *Overlapping* means that the estimates start every month, while *Nonoverlapping* means that until one monthly investment period (from 1 to 12 months) ends, the next one does not start. *n.a.* means not available.

Table 4: Percentiles of the betas calculated for each factor

<i>12m.US</i>	Observations	Months	Shares	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>Minimum</i>	390,429	247	3,285	-4.815	-11.305	-14.025	-14.576	-10.335	-13.374	-11.457	-3.077
25%				0.379	-0.594	-1.882	-1.373	-0.558	-1.399	-1.880	0.695
50%				0.940	0.020	-0.917	-0.470	-0.063	-0.596	-1.057	1.001
75%				1.626	0.656	-0.203	0.315	0.459	-0.004	-0.433	1.330
<i>Maximum</i>	10.835	7.252	9.029	14.768	8.195	15.154	5.826	5.261			
<i>60m.US</i>	Observations	Months	Shares	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>Minimum</i>	303,847	199	3,285	-1.585	-5.367	-7.225	-7.758	-2.903	-4.518	-6.805	-0.869
25%				0.433	-0.124	-1.998	-1.010	-0.522	-1.151	-1.779	0.753
50%				0.908	0.322	-1.146	-0.396	-0.221	-0.512	-1.085	1.011
75%				1.385	0.809	-0.437	0.074	0.011	-0.082	-0.542	1.257
<i>Maximum</i>	4.177	4.442	2.903	5.978	2.970	3.317	2.165	3.465			
<i>12m.Develop.ex - US</i>	Observations	Months	Shares	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>Minimum</i>	259,262	247	1,257	-6.571	-14.037	-6.773	-19.185	-21.553	-6.519	-18.377	-3.719
25%				-1.429	-0.482	-0.724	-1.899	-1.495	-1.272	-2.782	0.686
50%				-0.604	0.166	-0.062	-0.644	-0.397	-0.683	-1.781	0.965
75%				0.080	0.965	0.616	0.354	0.450	-0.190	-0.972	1.286
<i>Maximum</i>	7.600	10.995	10.982	16.408	11.562	4.627	7.549	4.404			
<i>60m.Develop.ex - US</i>	Observations	Months	Shares	SMB	HML	RMW	CMA	WML	BAB	QMJ	Mkt
<i>Minimum</i>	209,874	199	1,257	-4.976	-7.784	-4.121	-8.037	-8.862	-4.428	-10.937	0.000
25%				-1.628	-0.097	-0.847	-2.041	-1.413	-1.304	-2.756	0.752
50%				-0.852	0.354	-0.462	-0.908	-0.551	-0.836	-1.972	0.983
75%				-0.077	0.936	-0.136	-0.171	0.146	-0.424	-1.263	1.265
<i>Maximum</i>	3.826	6.529	3.860	4.745	4.708	1.428	1.011	3.175			

Note: *US* represents the factors for the U.S. market; *Develop.ex - US* represents the factors for the developed countries market excluding the U.S.; betas are calculate using daily data from 12 and 60 months (*m.*). *Observations* is size of sample, *Months* is the total number of months in which the beta has been calculated and *Shares* is the number of assets for which the beta has been calculated in one of the months of the sample.

Table 5: Fama-MacBeth risk premium estimation with calculated betas from 12 months of daily data

<i>1m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	-0.0001	0.0006	-0.0028	0.0008	0.0010	-0.0002	0.0004	0.0091	0.0075
<i>tprob</i>	0.3981	0.3612	0.2609	0.3535	0.2716	0.3982	0.3956	0.0141	0.0637
<i>R²</i>					6.13%				
<i>N</i>					247				
<i>m - M</i>					859-3036				
<i>Signif.</i>	168	147	144	130	114	126	144	155	143
<i>3m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0010	0.0000	-0.0086	0.0012	0.0043	-0.0067	-0.0003	0.0295	0.0144
<i>tprob</i>	0.3952	0.3985	0.1984	0.3843	0.1414	0.3509	0.3983	0.0073	0.1980
<i>R²</i>					6.78%				
<i>N</i>					244				
<i>m - M</i>					859-3036				
<i>Signif.</i>	185	125	145	128	108	144	151	149	153
<i>6m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0056	-0.0043	-0.0159	-0.0006	0.0082	-0.0302	-0.0033	0.0585	0.0223
<i>tprob</i>	0.3701	0.3104	0.1657	0.3976	0.0841	0.1473	0.3914	0.0057	0.2246
<i>R²</i>					7.41%				
<i>N</i>					241				
<i>m - M</i>					859-3036				
<i>Signif.</i>	182	146	144	124	102	128	157	166	156
<i>12m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0115	-0.0162	-0.0184	0.0029	0.0136	-0.0448	0.0037	0.1022	0.0592
<i>tprob</i>	0.3585	0.0958	0.2664	0.3863	0.0373	0.1837	0.3934	0.0005	0.0130
<i>R²</i>					8.07%				
<i>N</i>					235				
<i>m - M</i>					859-3036				
<i>Signif.</i>	200	129	118	118	118	141	137	154	152
<i>1m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0026	-0.0004	-0.0014	0.0044	-0.0003	0.0008	0.0054	-0.0030	-0.0008
<i>tprob</i>	0.2516	0.3910	0.3282	0.1701	0.3914	0.2950	0.0475	0.1941	0.3936
<i>R²</i>					13.41%				
<i>N</i>					772-1246				
<i>m - M</i>					247				
<i>Signif.</i>	157	143	119	142	136	117	138	128	121
<i>3m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0094	-0.0012	-0.0047	0.0146	0.0002	0.0031	0.0157	-0.0112	-0.0083
<i>tprob</i>	0.1858	0.3882	0.2522	0.0774	0.3976	0.2025	0.0416	0.1164	0.3353
<i>R²</i>					13.57%				
<i>N</i>					244				
<i>m - M</i>					772-1246				
<i>Signif.</i>	159	142	129	148	140	113	131	126	121
<i>6m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0230	-0.0040	-0.0075	0.0233	0.0001	0.0075	0.0337	-0.0252	-0.0312
<i>tprob</i>	0.0971	0.3583	0.2280	0.0688	0.3985	0.1098	0.0198	0.0442	0.1726
<i>R²</i>					13.31%				
<i>N</i>					241				
<i>m - M</i>					772-1246				
<i>Signif.</i>	169	137	113	135	127	123	117	111	117
<i>12m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0488	-0.0076	-0.0187	0.0272	0.0043	0.0084	0.0669	-0.0324	-0.0575
<i>tprob</i>	0.0352	0.3345	0.0462	0.1171	0.3605	0.2035	0.0020	0.0720	0.1145
<i>R²</i>					13.02%				
<i>N</i>					235				
<i>m - M</i>					772-1246				
<i>Signif.</i>	178	124	94	111	132	114	114	107	115

Note: *US* represents the factors for the U.S. market; *DEV.* represents the factors for the developed countries market excluding the U.S.; betas are calculate using daily data from 12 months (*m.*); *Param.* is the estimation for each risk premium; *return* is the dependent variable, it is estimated as the expected excess return on risk-free rate for each investment term (from 1 to 12 months); *tprob* is the statistical significance of each parameter (risk premium) robust to heteroscedasticity and autoregression (Newey-West with an optimum of lags for each case); *R²* is the mean of the determination coefficient of the cross-sectional estimates; *N* is the number estimation in cross-sectional; *m - M* are the number of companies for cross-sectional estimate minimum and maximum, respectively; *Signif.* is the number of times it is significant at 5% in cross-sectional estimates each factor.

Table 6: Fama-MacBeth risk premium estimation with calculated betas from 60 months of daily data

<i>1m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	-0.0034	-0.0021	-0.0029	0.0030	0.0016	0.0004	-0.0019	0.0108	0.0174
<i>tprob</i>	0.2357	0.2271	0.3370	0.1980	0.2087	0.3974	0.3608	0.0456	0.0000
<i>R²</i>					6.52%				
<i>N</i>					198				
<i>m - M</i>					1013-2321				
<i>Signif.</i>					96	101	101	128	102
<i>3m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	-0.0091	-0.0058	-0.0122	0.0081	0.0048	-0.0074	-0.0066	0.0334	0.0504
<i>tprob</i>	0.2428	0.2087	0.2559	0.1774	0.1737	0.3482	0.3381	0.0247	0.0000
<i>R²</i>					7.34%				
<i>N</i>					196				
<i>m - M</i>					1013-2321				
<i>Signif.</i>					95	101	109	135	117
<i>6m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	-0.0162	-0.0118	-0.0203	0.0122	0.0085	-0.0253	-0.0131	0.0709	0.0994
<i>tprob</i>	0.2475	0.1478	0.2537	0.1911	0.1641	0.2078	0.3214	0.0052	0.0000
<i>R²</i>					8.46%				
<i>N</i>					193				
<i>m - M</i>					1013-2321				
<i>Signif.</i>					96	101	116	155	119
<i>12m.US</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	-0.0203	-0.0245	-0.0224	0.0182	0.0161	-0.0597	-0.0401	0.1564	0.1849
<i>tprob</i>	0.3126	0.0692	0.3190	0.1075	0.0817	0.0583	0.1806	0.0001	0.0000
<i>R²</i>					9.96%				
<i>N</i>					187				
<i>m - M</i>					853-1208				
<i>Signif.</i>					102	98	130	167	132
<i>1m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0021	-0.0019	-0.0049	0.0000	0.0021	0.0001	0.0047	-0.0029	-0.0075
<i>tprob</i>	0.3352	0.2642	0.0255	0.3984	0.2302	0.3980	0.1876	0.2623	0.2224
<i>R²</i>					14.27%				
<i>N</i>					198				
<i>m - M</i>					198				
<i>Signif.</i>					116	105	127	123	111
<i>3m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0070	-0.0056	-0.0178	-0.0022	0.0073	0.0004	0.0135	-0.0103	-0.0261
<i>tprob</i>	0.3075	0.2488	0.0021	0.3925	0.1660	0.3969	0.1793	0.1865	0.1515
<i>R²</i>					14.76%				
<i>N</i>					196				
<i>m - M</i>					853-1208				
<i>Signif.</i>					118	98	122	120	112
<i>6m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0147	-0.0094	-0.0367	-0.0050	0.0137	0.0010	0.0231	-0.0231	-0.0600
<i>tprob</i>	0.2815	0.2541	0.0005	0.3864	0.1409	0.3947	0.1911	0.1005	0.0867
<i>R²</i>					14.94%				
<i>N</i>					193				
<i>m - M</i>					853-1208				
<i>Signif.</i>					121	94	114	104	113
<i>12m.DEV.</i>	<i>Const</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>Param.</i>	0.0345	-0.0159	-0.0709	-0.0028	0.0294	0.0093	0.0324	-0.0567	-0.1516
<i>tprob</i>	0.2005	0.2512	0.0004	0.3962	0.0733	0.3065	0.1940	0.0112	0.0215
<i>R²</i>					15.54%				
<i>N</i>					187				
<i>m - M</i>					853-1208				
<i>Signif.</i>					115	97	104	106	130

Note: *US* represents the factors for the U.S. market; *DEV.* represents the factors for the developed countries market excluding the U.S.; betas are calculate using daily data from 60 months (*m.*); *Param.* is the estimation for each risk premium; *return* is the dependent variable, it is estimated as the expected excess return on risk-free rate for each investment term (from 1 to 12 months); *tprob* is the statistical significance of each parameter (risk premium) robust to heteroscedasticity and autoregression (Newey-West with an optimum of lags for each case); *R²* is the mean of the determination coefficient of the cross-sectional estimates; *N* is the number estimation in cross-sectional; *m - M* are the number of companies for cross-sectional estimate minimum and maximum, respectively; *Signif.* is the number of times it is significant at 5% in cross-sectional estimates each factor.

Table 7: Sharpe ratio of extreme portfolios for each factor

<i>T</i>	<i>P</i>	<i>Betas12m.US</i>								<i>Betas60m.US</i>							
		<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1	5%	-0.066	-0.082	-0.087	-0.114	-0.132	-0.003	-0.032	-0.061	-0.053	-0.094	-0.110	-0.163	-0.136	0.035*	-0.039	-0.042
	95%	-0.051	-0.018	-0.001	-0.015	0.027*	-0.104	-0.117	-0.052	-0.134	-0.018	-0.023	0.063*	0.027*	-0.123	-0.132	-0.012
3	5%	-0.091	-0.135	-0.136	-0.185	-0.217	0.013*	-0.030	-0.062	-0.105	-0.158	-0.170	-0.265	-0.222	0.050*	-0.069	-0.066
	95%	-0.074	-0.001	0.013*	0.043*	0.067*	-0.150	-0.168	-0.080	-0.205	-0.024	-0.039	0.115*	0.051*	-0.219	-0.183	-0.031
6	5%	-0.112	-0.159	-0.189	-0.234	-0.283	0.029*	-0.017	-0.043	-0.132	-0.215	-0.227	-0.324	-0.279	0.075*	-0.080	-0.090
	95%	-0.105	-0.005	0.030*	0.100*	0.106*	-0.180	-0.219	-0.118	-0.236	-0.015	-0.039	0.186*	0.076*	-0.292	-0.227	-0.046
12	5%	-0.182	-0.204	-0.243	-0.318	-0.361	0.028*	-0.030	-0.067	-0.165	-0.284	-0.269	-0.358	-0.332	0.164*	-0.105	-0.113
	95%	-0.158	-0.101	0.032*	0.138*	0.144*	-0.239	-0.300	-0.179	-0.257	-0.034	-0.041	0.334*	0.103*	-0.355	-0.279	-0.080
<i>T</i>	<i>P</i>	<i>Betas12m.DEV.ex-US</i>								<i>Betas60m.DEV.ex-US</i>							
		<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1	5%	-0.001	0.055*	-0.005	0.007*	0.000	0.021*	0.019*	0.103*	-0.026	-0.042	-0.032	-0.029	-0.027	-0.019	-0.047	0.006*
	95%	0.127*	0.035*	0.036	0.085*	0.019*	0.005*	0.076*	0.042*	0.008*	-0.003	-0.033	-0.013	-0.042	-0.047	0.009*	-0.021
3	5%	0.010*	0.079*	-0.001	0.019*	0.014*	0.036*	0.025*	0.195*	-0.043	-0.062	-0.044	-0.044	-0.040	-0.027	-0.082	0.019*
	95%	0.194*	0.052*	0.047*	0.169*	0.040*	0.013*	0.131*	0.063*	0.019*	-0.014	-0.060	-0.024	-0.054	-0.088	0.027*	-0.039
6	5%	0.035*	0.115*	0.006*	0.024*	0.042*	0.053*	0.027*	0.297*	-0.054	-0.076	-0.050	-0.052	-0.045	-0.042	-0.112	0.043*
	95%	0.275*	0.065*	0.068*	0.252*	0.051*	0.042*	0.169*	0.083*	0.019*	-0.036	-0.079	-0.029	-0.049	-0.108	0.037*	-0.050
12	5%	0.033*	0.107*	0.000	-0.004	0.122*	0.047*	0.003*	0.413*	-0.083	-0.120	-0.056	-0.112	-0.081	-0.091	-0.192	0.073*
	95%	0.359*	0.048*	0.097*	0.303*	0.034*	0.097*	0.237*	0.084*	0.011*	-0.095	-0.135	-0.070	-0.066	-0.152	0.025*	-0.093

Note: *T* represents the investment period in months (*m.*); *P* shows the percentile of betas for each factor used to include an asset in the extreme portfolio factor; * shows the Sharpe ratio positive.

Table 8: Sharpe ratio of conditional factor investing

<i>US 12m.Betas</i>									
<i>T</i>	<i>P</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1m.	5%	-0.086	0.025	-0.173	-0.173	-0.107	-0.034	0.070	-0.138
	95%	-0.082	0.122	-0.086	-0.038	0.063	-0.130	-0.059	-0.028
3m.	5%	-0.057	-0.112	-0.170	-0.281	-0.123	0.040	0.014	-0.100
	95%	-0.036	0.104	-0.094	0.005	0.123	-0.208	-0.139	-0.096
6m.	5%	-0.058	-0.160	-0.176	-0.265	-0.208	0.041 *	0.020	-0.163
	95%	-0.046	-0.045	-0.002	0.141 *	0.264 *	-0.180	-0.176	-0.120
12m.	5%	-0.091	-0.067	-0.346	-0.480	-0.380	0.016	-0.010	-0.170
	95%	-0.192	0.174 *	-0.139	0.009	0.164	-0.285	-0.320	-0.197
<i>Development ex - US 12m.Betas</i>									
<i>T</i>	<i>P</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1m.	5%	-0.017	0.099	-0.059	-0.087	0.019	0.049	0.029	0.184
	95%	0.142	0.102	0.068	-0.023	0.023	0.019	0.133	0.080
3m.	5%	0.078	0.112	-0.090	0.138	0.000	0.129 *	-0.077	0.249
	95%	0.230	0.152	0.014	0.178	0.010	0.141	0.106	0.057
6m.	5%	0.128	0.015	0.023	0.049	0.041	0.097	-0.081	0.293
	95%	0.385	0.111	0.069 *	0.288 *	0.173 *	0.052	0.086	0.098
12m.	5%	0.275	0.061	-0.084	0.005	0.150	0.015	0.052	0.693 *
	95%	0.722 *	0.907 *	-0.194	0.255	0.073	0.084	0.409 *	0.336
<i>US 12m.Betas</i>									
<i>T</i>	<i>P</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1m.	5%(-)	-0.143	0.245 *	0.096	0.536	0.015	-0.126	0.350 *	0.199
	95%(+)	0.296	0.093	0.041	-0.278	0.319	0.158 *	0.015	-0.077
3m.	5%(-)	-0.253	0.167	0.253 *	1.019	-0.113	-0.272	0.125	0.404 *
	95%(+)	0.561	0.141	0.080	-0.297	0.465	0.145	-0.118	-0.357
6m.	5%(-)	-0.297	0.096	0.122	0.907	-0.202	-0.249	0.064	0.337
	95%(+)	0.608 *	0.054	0.068	-0.167	0.481 *	0.126	-0.228	-0.363
12m.	5%(-)	-0.424	0.221	-0.243	1.563 *	-0.220	-0.374	0.118	0.395
	95%(+)	0.581	0.162	0.218	-0.314	0.207	-0.008	-0.574	-0.463
<i>Development ex - US 12m.Betas</i>									
<i>T</i>	<i>P</i>	<i>BAB</i>	<i>CMA</i>	<i>HML</i>	<i>QMJ</i>	<i>RMW</i>	<i>SMB</i>	<i>WML</i>	<i>Mkt</i>
1m.	5%(-)	-0.223	0.514	0.253	0.494	0.450	0.339	-0.016	0.373
	95%(+)	0.738	0.367	0.114	-0.331	0.406	0.118	0.440	-0.023
3m.	5%(-)	-0.026	0.731	0.151	0.952 *	0.443	0.353 *	-0.170	0.485
	95%(+)	0.589	0.309	0.051	-0.297	0.236	0.297	0.633	-0.159
6m.	5%(-)	-0.068	0.650	0.061	0.733	0.456	0.249	-0.162	0.693
	95%(+)	0.730	0.101	0.326	-0.144	0.334	0.139	0.588	-0.081
12m.	5%(-)	-0.006	0.956 *	-0.132	0.633	0.931 *	0.162	0.052	0.867 *
	95%(+)	1.029 *	-0.165	0.551 *	-0.175	0.044	0.156	0.922 *	0.218

Note: *T* represents the investment period in months (m.); *P* shows the percentile of betas for each factor used to include an asset in the extreme portfolio factor, i.e., 5% represents the investment in assets with the lowest betas with respect to the factor, located up to the 5% percentile at each time instant; 95% refers to the investment in assets with the highest betas with respect to the factor located at the 95% percentile and above; (+) means that the risk premium was positive and significantly different from zero, while (-) represents that it was significant and negative; * represents the highest remuneration per unit of risk.

Table 9: Sharpe ratio of conditional factor investing non-overlapping

<i>B</i>	<i>T</i>	<i>Sp</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>	<i>WML</i>	<i>BAB</i>	<i>QMJ</i>	<i>Mkt</i>
<i>US Factor investing strategies</i>										<i>Development ex - US Factor investing strategies</i>								
12m.	1m.	S	0.028	-0.033	0.126	0.034	-0.021	0.172	0.092	0.194	-0.034	0.090	0.141	0.077	0.193	0.423	0.216	0.127
		p	245	245	245	245	245	245	245	245	245	245	245	245	245	245	245	245
	3m.	S	0.049	-0.043	0.189	0.058	-0.037	0.278	0.130	0.309	-0.070	0.137	0.218	0.101	0.335	0.672	0.319	0.205
		p	81	81	81	81	81	81	81	81	81	81	81	81	81	81	81	81
	6m.	S	0.048	-0.063	0.381	0.062	-0.017	0.351	0.254	0.437	-0.111	0.167	0.341	0.108	0.465	0.850	0.517	0.250
		p	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40
	12m.	S	-0.006	-0.028	0.519	0.117	0.019	0.474	0.395	0.487	-0.116	0.213	0.453	0.126	0.806	1.139	0.804	0.281
		p	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19	19
	1m.	S	0.007	0.079	0.169	0.022	0.019	0.099	0.144	0.156	-0.082	0.035	0.134	0.077	0.234	0.367	0.287	0.056
		p	197	197	197	197	197	197	197	197	197	197	197	197	197	197	197	197
	3m.	S	-0.018	-0.115	0.266	0.040	-0.034	0.167	0.223	0.250	-0.160	0.053	0.220	0.101	0.424	0.586	0.461	0.084
		p	65	65	65	65	65	65	65	65	65	65	65	65	65	65	65	65
60m.	6m.	S	-0.022	-0.150	0.413	0.055	-0.049	0.192	0.323	0.361	-0.216	0.054	0.276	0.107	0.574	0.698	0.635	0.101
		p	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32
	12m.	S	0.042	-0.068	0.466	0.228	-0.083	0.371	0.413	0.426	-0.128	0.092	0.311	0.125	0.867	0.983	0.832	0.116
		p	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15
12m.	1m.	S	0.208	0.271	0.253	0.336	0.247	0.176	0.183	0.206	0.390	0.364	0.409	0.336	0.269	0.314	0.136	0.199
		p	158	148	145	131	143	143	169	148	142	117	141	137	117	138	127	121
	3m.	S	0.148	0.238	0.237	0.266	0.288	0.209	0.148	0.161	0.288	0.379	0.420	0.203	0.059	0.004	0.280	0.076
		p	41	56	56	45	46	47	53	54	47	43	54	52	39	46	40	40
	6m.	S	0.049	0.307	0.156	0.084	0.063	0.021	0.369	0.011	0.337	0.037	0.556	0.445	0.502	0.157	0.927	0.012
		p	20	25	25	23	24	25	27	23	24	19	22	21	24	19	19	19
	12m.	S	0.041	0.664	0.014	0.020	0.022	0.025	0.646	0.042	0.066	0.148	2.777	0.516	1.124	0.335	0.411	0.048
		p	13	9	13	13	13	14	6	6	10	10	6	9	10	10	13	13
	1m.	S	0.305	0.219	0.270	0.133	0.257	0.109	0.138	0.125	0.286	0.450	0.084	0.144	0.136	0.106	0.072	0.138
		p	131	120	121	101	118	120	128	112	122	99	119	117	102	127	124	109
	3m.	S	0.040	0.286	0.243	0.051	0.122	0.015	0.114	0.100	0.297	0.210	0.031	0.169	0.098	0.066	0.347	0.012
		p	44	45	43	37	43	31	46	36	40	29	38	37	30	39	40	44
60m.	6m.	S	0.198	0.012	0.027	0.257	0.125	0.048	0.866	0.150	0.075	0.126	0.021	0.584	0.041	0.175	0.354	0.018
		p	32	20	24	18	24	20	24	21	22	15	17	22	17	19	18	22
	12m.	S	0.045	0.057	0.094	0.010	0.002	0.017	0.699	0.010	0.335	0.104	0.416	0.300	0.477	0.021	0.796	0.290
		p	11	9	10	9	8	12	10	8	9	8	8	9	7	8	9	10

Note: *B* shows the sample size in months (m.) of daily data used to calculate the betas; *T* represents the investment period in months (m.); *Sp* shows the Sharpe ratio for each strategy or factor (*S*) and the number of periods (*p*) of the selected investment size (*T*) in which the investment is carried out.

Table 10: Companies and participation in factor investing strategies

<i>Factor + Term</i>	<i>US 12m.</i>			<i>US 60m.</i>		
	<i>Total</i>	<i>Asset</i>	<i>Max.%</i>	<i>Total</i>	<i>Asset</i>	<i>Max.%</i>
<i>BAB + 1m.</i>	1865	Patterson-UTI Energy Inc	0.40%	900	Century Aluminum Co	0.73%
<i>BAB + 3m.</i>	1347	Century Aluminum Co	0.54%	556	Century Aluminum Co	0.81%
<i>BAB + 6m.</i>	1055	Century Aluminum Co	0.46%	463	Century Aluminum Co	0.84%
<i>BAB + 12m.</i>	595	APA Corp.	0.71%	379	Century Aluminum Co	1.13%
<i>CMA + 1m.</i>	1680	Century Aluminum Co	0.45%	880	Hess Corp	0.54%
<i>CMA + 3m.</i>	1235	Micron Technology Inc	0.45%	727	EOG Resources Inc	0.47%
<i>CMA + 6m.</i>	1041	Century Aluminum Co	0.45%	551	EOG Resources Inc	0.53%
<i>CMA + 12m.</i>	597	Altria Group Inc	0.57%	435	Bancorp Inc	0.62%
<i>HML + 1m.</i>	1611	Zions Bancorporation NA	0.54%	820	Huntington Bancshares Inc	0.76%
<i>HML + 3m.</i>	1279	Zions Bancorporation NA	0.57%	694	Huntington Bancshares Inc	0.78%
<i>HML + 6m.</i>	972	SSR Mining Inc	0.47%	593	Huntington Bancshares Inc	0.63%
<i>HML + 12m.</i>	595	Century Aluminum Co	0.63%	478	Huntington Bancshares Inc	0.59%
<i>QMJ + 1m.</i>	1784	Patterson-UTI Energy Inc	0.63%	781	General Mills Inc	0.97%
<i>QMJ + 3m.</i>	1297	Patterson-UTI Energy Inc	0.75%	662	General Mills Inc	1.10%
<i>QMJ + 6m.</i>	963	Clorox Co	0.70%	562	General Mills Inc	1.07%
<i>QMJ + 12m.</i>	668	Clorox Co	0.90%	341	General Mills Inc	1.28%
<i>RMW + 1m.</i>	1686	Altria Group Inc	0.49%	803	Consolidated Edison Inc	0.68%
<i>RMW + 3m.</i>	1365	Altria Group Inc	0.47%	716	Consolidated Edison Inc	0.55%
<i>RMW + 6m.</i>	943	Southern Co	0.42%	633	Sangamo Therapeutics Inc	0.69%
<i>RMW + 12m.</i>	629	Patterson-UTI Energy Inc	0.50%	401	Consolidated Edison Inc	0.76%
<i>SMB + 1m.</i>	1694	Patterson-UTI Energy Inc	0.57%	704	FirstEnergy Corp	0.92%
<i>SMB + 3m.</i>	1173	Procter & Gamble Co	0.56%	592	FirstEnergy Corp	1.06%
<i>SMB + 6m.</i>	712	Southern Co	0.82%	468	FirstEnergy Corp	1.07%
<i>SMB + 12m.</i>	663	Procter & Gamble Co	0.75%	361	FirstEnergy Corp	1.05%
<i>WML + 1m.</i>	1815	Century Aluminum Co	0.58%	986	United Airlines Holdings Inc	0.65%
<i>WML + 3m.</i>	1369	Century Aluminum Co	0.50%	772	Century Aluminum Co	0.69%
<i>WML + 6m.</i>	983	Century Aluminum Co	0.45%	625	Century Aluminum Co	0.85%
<i>WML + 12m.</i>	701	Southern Co	0.54%	387	ServiceNow Inc	0.88%
<i>Mkt + 1m.</i>	1684	Century Aluminum Co	0.70%	838	ServiceNow Inc	0.93%
<i>Mkt + 3m.</i>	1318	Union Bankshares Inc	0.70%	640	ServiceNow Inc	0.97%
<i>Mkt + 6m.</i>	940	Century Aluminum Co	0.78%	481	ServiceNow Inc	1.02%
<i>Mkt + 12m.</i>	330	Century Aluminum Co	1.14%	337	ServiceNow Inc	1.10%
<i>Factor + Term</i>	<i>Develop. ex - US 12m.</i>			<i>Develop. ex - US 60m.</i>		
	<i>Total</i>	<i>Asset</i>	<i>Max.%</i>	<i>Total</i>	<i>Asset</i>	<i>Max.%</i>
<i>BAB + 1m.</i>	880	Kinross Gold Corp	0.62%	426	Commerzbank AG	0.96%
<i>BAB + 3m.</i>	732	Kinross Gold Corp	0.75%	380	Tourism Holdings Ltd	0.97%
<i>BAB + 6m.</i>	517	Kinross Gold Corp	0.80%	312	Tourism Holdings Ltd	1.13%
<i>BAB + 12m.</i>	345	Kinross Gold Corp	0.97%	226	Tourism Holdings Ltd	1.38%
<i>CMA + 1m.</i>	913	First Majestic Silver Corp	0.56%	537	PGS ASA	0.87%
<i>CMA + 3m.</i>	789	First Majestic Silver Corp	0.57%	436	Shiseido Co Ltd	0.82%
<i>CMA + 6m.</i>	550	Teck Resources Ltd	0.87%	413	PGS ASA	0.90%
<i>CMA + 12m.</i>	420	Sartorius AG	0.73%	296	PGS ASA	0.97%
<i>HML + 1m.</i>	852	M3 Inc	0.64%	492	CyberAgent Inc	1.07%
<i>HML + 3m.</i>	704	M3 Inc	0.81%	362	CyberAgent Inc	1.09%
<i>HML + 6m.</i>	545	M3 Inc	0.76%	302	CyberAgent Inc	1.50%
<i>HML + 12m.</i>	375	Agnico Eagle Mines Ltd	1.05%	256	CyberAgent Inc	1.41%
<i>QMJ + 1m.</i>	897	Commerzbank AG	0.76%	424	Commerzbank AG	1.08%
<i>QMJ + 3m.</i>	654	Commerzbank AG	0.89%	362	Commerzbank AG	1.10%
<i>QMJ + 6m.</i>	510	Commerzbank AG	0.87%	313	Commerzbank AG	1.37%
<i>QMJ + 12m.</i>	382	Commerzbank AG	1.03%	190	Commerzbank AG	1.44%
<i>RMW + 1m.</i>	909	MAG Silver Corp	0.50%	487	Barclays PLC	0.95%
<i>RMW + 3m.</i>	767	First Quantum Minerals Ltd	0.57%	436	Barclays PLC	0.97%
<i>RMW + 6m.</i>	526	First Quantum Minerals Ltd	0.66%	322	Barclays PLC	0.98%
<i>RMW + 12m.</i>	287	Anglo American PLC	0.86%	271	CyberAgent Inc	1.07%
<i>SMB + 1m.</i>	729	Deutsche Bank AG	0.84%	397	Barclays PLC	1.09%
<i>SMB + 3m.</i>	592	Deutsche Bank AG	1.02%	330	Barclays PLC	0.93%
<i>SMB + 6m.</i>	494	Deutsche Bank AG	1.07%	308	CyberAgent Inc	0.97%
<i>SMB + 12m.</i>	377	Deutsche Bank AG	1.08%	265	Barclays PLC	1.11%
<i>WML + 1m.</i>	931	Commerzbank AG	0.64%	517	Baytex Energy Corp	0.91%
<i>WML + 3m.</i>	752	Baytex Energy Corp	0.73%	425	Baytex Energy Corp	1.12%
<i>WML + 6m.</i>	628	Commerzbank AG	0.83%	306	Baytex Energy Corp	1.26%
<i>WML + 12m.</i>	348	Commerzbank AG	0.79%	193	Baytex Energy Corp	1.34%
<i>Mkt + 1m.</i>	865	CLP Holdings Ltd	0.83%	419	Power Assets Holdings Ltd	1.05%
<i>Mkt + 3m.</i>	693	CLP Holdings Ltd	0.74%	359	Power Assets Holdings Ltd	1.21%
<i>Mkt + 6m.</i>	520	CLP Holdings Ltd	0.95%	343	Power Assets Holdings Ltd	1.23%
<i>Mkt + 12m.</i>	473	Power Assets Holdings Ltd	0.94%	252	Power Assets Holdings Ltd	1.58%

Note: *Factor + Term* represents the factor (*Factor*) on which the conditional strategy is performed and the term (*Term*) of the non-overlapping investment in months (*m.*). *Total* shows the total number of companies participating at least once in the conditional strategy of the corresponding factor and time frame. *Asset* is the name of the company that participates more often (*Max.%*) in the corresponding conditional strategy.

Table 11: Sharpe ratio of combined conditional factor investing

<i>B</i>	<i>T</i>	<i>US Factors</i>			<i>Development ex - US Factors</i>		
		<i>CCe</i>	<i>BAB</i>	<i>Mkt</i>	<i>CCe</i>	<i>BAB</i>	<i>Mkt</i>
12m.	1m.	0.335	0.172	0.194	0.442	0.423	0.127
	3m.	0.365	0.278	0.309	0.346	0.672	0.205
	6m.	0.501	0.351	0.437	0.511	0.850	0.250
	12m.	0.129	0.474	0.487	0.184	1.139	0.281
60m.	1m.	0.333	0.099	0.156	0.293	0.367	0.056
	3m.	0.272	0.167	0.250	0.232	0.586	0.084
	6m.	0.226	0.192	0.361	0.216	0.698	0.101
	12m.	0.133	0.371	0.426	0.177	0.983	0.116

Note: *B* shows the sample size in months (*m.*) of daily data used to calculate the betas; *T* represents the investment period in months (*m.*); *CCe* shows the

Sharpe ratio for combined conditional factor investing strategy.